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Multimodal generative cyber forensics network for unified malware intelligence synthesis through temporal evidence alignment, neuro-symbolic hypergraph reasoning, and cognitive threat attributions

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Abstract

Cyber-forensic investigations increasingly involve heterogeneous evidence, including phishing audio, malicious text and scripts, executable binaries, memory artifacts, and network traffic. Existing approaches generally analyze these sources independently or perform sample-level fusion without explicitly addressing temporal asynchrony, higher-order forensic relationships, missing attack stages, counterfactual threat evolution, and attribution uncertainty. This paper presents a Multimodal Generative Cyber Forensics Network comprising CTEEN for temporal-semantic alignment, NSFRH for higher-order forensic reasoning, GASRE for evidence-constrained attack-stage reconstruction, CMES for counterfactual malware-evolution analysis, and CTAECO for confidence-aware candidate-category attribution. The evaluation used VirusShare, CIC-MalMem-2022, the Android Malware Corpus, the Malware Traffic Repository, and an Integrated Controlled Multimodal Benchmark constructed from source pools containing approximately 52,000 malware records, 1.8 million network-flow records, 24,000 textual or script artifacts, and 12,500 phishing or social-engineering audio recordings. Comparisons were conducted against Cyber Code Intelligence, AI-Based Malware Analysis, Multimodal Pre-trained Fusion, and LLM-MalDetect using identical eligible evidence groups, preprocessing outputs, partitions, and metric definitions. Across 25 matched partition–seed observations, the proposed framework achieved 99.26±0.18% classification accuracy (95% CI: 99.19–99.33%), 99.20±0.21% F1-score, and 99.66±0.12% AUC. It further obtained 98.46% mean controlled alignment accuracy, 98.68% evidence-constrained reconstruction accuracy, 99.04% benchmark-defined candidate-category attribution accuracy, and 99.0% overall forensic-intelligence reliability. Accuracy decreased to 95.18% under temporally disjoint testing and 92.34% under family-disjoint testing. Leakage audits, negative controls, matched statistical tests, and alternative-design ablations further supported the findings. The complete framework contained 96.3 million trainable parameters and required 12.8 GB of peak memory, with a mean inference latency of 73.2 ms per complete multimodal episode (13.7 episodes/s) under the reported computational environment. These requirements indicate that deployment on resource-constrained platforms would require model compression or inference optimization. Reviewer 3, Comment 3) The results demonstrate effective evidence-constrained multimodal reasoning under

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the controlled benchmark but do not establish natural incident-level correspondence, definitive real-world attribution, or universal forensic performance.

Keywords Multimodal cyber forensics · Malware intelligence synthesis · Hypergraph threat reasoning · Generative attack reconstruction · Cognitive threat attribution · Analysis

Abbreviations

MGCFN	Multimodal generative cyber forensics network
CTEEN	Cross-modal temporal evidence encoding network
NSFRH	Neuro-symbolic forensic relationship hypergraph
GASRE	Generative attack-sequence reconstruction engine
CMES	Counterfactual malware-evolution simulator
CTAECO	Cognitive threat attribution and evidence confidence optimizer
UTEM	Unified temporal evidence matrix
FHKS	Forensic hypergraph knowledge structure

Introduction

Modern cyberattacks generate heterogeneous evidence across communication, host, executable, memory, and network environments. A single investigation may contain voice-based social-engineering or phishing recordings[1,2–3], malicious scripts, executable binaries[4,5,6], system logs, and network-traffic traces. These artifacts differ in structure, sampling frequency, semantic granularity, and observation time. Their joint interpretation is therefore substantially more difficult than conventional malware analysis based on a single evidence source.

Contemporary threat actors frequently conduct coordinated campaigns across multiple communication channels and execution environments. Evidence belonging to such campaigns may be distributed across emails, voice communication, scripts, executable files, host activity, and network connections. Conventional forensic tools generally examine these sources independently, producing fragmented findings that are difficult to combine into a traceable investigation sequence. The growing use of stealth, obfuscation, polymorphism, and multi-stage execution has made reliable evidence correlation increasingly important for cybersecurity investigation and incident response[7,8–9].

Most existing malware-forensic methods focus on a specific task or evidence modality, such as executable analysis, malware classification, network-intrusion detection[10,11,12], or phishing-content inspection. Although these approaches provide useful modality-specific findings, they offer limited support for modelling the dependencies among heterogeneous artifacts. This limitation can prevent investigators from reconstructing an attack lifecycle, identifying relationships among separated stages, and estimating the confidence associated with a forensic interpretation[13,14–15].

Conventional multimodal-fusion approaches also commonly integrate static representations without explicitly resolving temporal asynchrony. A spoken phishing instruction may precede the delivery of a malicious message, execution of a script, activation of a binary, and generation of command-and-control traffic by different time intervals. Exact timestamp matching can miss meaningful relationships, whereas an excessively broad temporal window may associate unrelated artifacts. Multimodal cyber forensics consequently requires temporal-semantic correspondence rather than simple timestamp matching or direct feature concatenation.

This limitation can be observed in representative recent systems. Cyber Code Intelligence[3] learns semantic relationships from Android code but does not incorporate asynchronous communication, executable, and network evidence from a common investigation sequence. Multimodal Pre-trained Fusion[16] combines multiple Android-application representations, but its modalities describe different views of the same application rather

than temporally separated incident artifacts. Similarly, LLM-MalDetect[44] uses language-model-assisted representations for Android malware detection but does not reconstruct missing attack stages or estimate cross-modal forensic correspondence. These systems provide strong malware-classification capabilities within their intended scope; however, they do not address the temporal-semantic evidence-linkage problem investigated in this study.

The structure of forensic relationships presents another challenge. Conventional pairwise graphs connect two entities at a time, although one attack stage may be jointly supported by several evidence sources. For example, a phishing instruction, malicious command, opcode sequence, and outbound connection may collectively describe a controlled delivery-to-execution relationship. Representing this relationship through isolated pairwise links may fragment the shared context and weaken attack-stage interpretation.

Digital evidence is also frequently incomplete. Logs may be deleted, packet captures may be partial, communications may be encrypted, and intermediate attack stages may remain unobserved. Generative models can estimate missing information, but unrestricted generation may produce plausible statements that are not supported by the available forensic evidence. Reconstructed stages must therefore be constrained by observed relationships and distinguished from directly collected artifacts.

Counterfactual analysis introduces a related interpretive risk. Simulated variations can assist investigators in examining how malware behaviour might change under an altered payload, command server, persistence mechanism, communication delay, or obfuscation strategy. However, these outputs are prospective analytical scenarios rather than established historical events. Similarly, attribution should express uncertainty, remain limited to the candidate categories represented in the available evidence, and permit an inconclusive decision when forensic support is insufficient.

To address these challenges, this study presents a Multimodal Generative Cyber Forensics Network that connects temporal-semantic evidence alignment, neuro-symbolic hypergraph reasoning, evidence-constrained attack-stage reconstruction, counterfactual malware-evolution analysis, and confidence-aware candidate-category attribution. Unlike direct multimodal concatenation, the proposed framework maintains a traceable evidence flow and preserves the distinction among observed evidence, reconstructed historical-stage hypotheses, and prospective counterfactual projections.

As illustrated in Fig. 1, the proposed framework begins with phishing or social-engineering audio, malicious text and scripts, executable evidence, and network traffic. Modality-specific encoders preserve the characteristics of each evidence type before CTEEN estimates their temporal-semantic correspondence. The aligned representations are supplied to NSFRH, which constructs higher-order forensic relationships among multiple evidence entities. GASRE uses this evidence structure to estimate missing benchmark-defined attack stages while penalizing unsupported candidates. CMES generates prospective malware variations without treating them as observed or

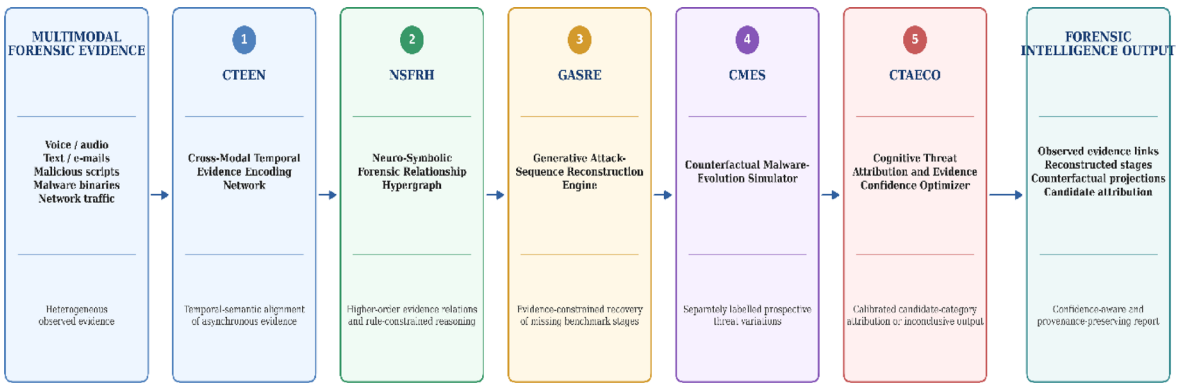


Fig. 1 End-to-end architecture of the proposed MGCFN, illustrating the provenance-preserving flow of controlled multimodal evidence through CTEEN-based temporal-semantic alignment, NSFRH-based higher-order forensic reasoning, GASRE-based evidence-constrained attack-stage reconstruction, CMES-based counterfactual malware-evolution analysis, and CTAECO-based confidence-calibrated candidate-category attribution.

reconstructed historical evidence. CTAECO subsequently integrates evidence consistency, symbolic-rule satisfaction, reconstruction support, and predictive uncertainty to produce confidence-aware candidate-category attribution or an inconclusive outcome.

The source repositories used in the evaluation were collected independently and do not provide naturally paired four-modality incidents. Accordingly, the relationships represented in Fig. 1 refer to controlled correspondence defined through the episode-construction protocol in Section IV. They should not be interpreted as proof that independently collected artifacts originated from the same operational incident.

Motivation

The primary motivation for this work is the widening gap between isolated malware detection and complete multimodal forensic reasoning. Existing systems can identify malicious code, suspicious content, or abnormal network behaviour, but they generally provide limited support for correlating asynchronous artifacts, representing higher-order relationships, recovering missing attack stages, separating historical findings from prospective simulations, and reporting attribution uncertainty within one analytical framework.

A practical forensic system must answer more than whether an individual artifact is malicious. It must determine which evidence entities are meaningfully related, how several artifacts jointly support an attack stage, which stages may be missing, how an altered adversarial condition could influence the interpretation, and whether the available evidence is sufficiently reliable to support a candidate attribution.

The proposed framework is therefore motivated by the need for a provenance-preserving analyst-support system. Observed evidence must remain distinguishable from reconstructed stages, while reconstructed stages must remain distinguishable from counterfactual projections. Attribution must be accompanied by confidence and must remain inconclusive when evidence is insufficient.

A transparent evaluation is equally important. Controlled correspondence must be clearly distinguished from natural incident-level pairing, related artifacts must remain isolated across data partitions, and performance must be examined under temporal and malware-family distribution shifts. Run-to-run variability, leakage auditing, negative controls, confidence calibration, alternative-design ablation, and computational profiling are therefore included in the evaluation methodology.

Contributions

The first contribution is a Cross-Modal Temporal Evidence Encoding Network that aligns asynchronous audio, textual, binary, and network evidence using temporal-semantic correspondence. CTEEN avoids reliance on exact timestamp equality and does not assume that artifacts from independent repositories were naturally collected from the same incident.

The second contribution is a Neuro-Symbolic Forensic Relationship Hypergraph that represents higher-order relationships among several evidence entities. NSFRH combines learned hypergraph propagation with forensic-rule constraints, allowing voice intent, textual commands, executable behaviour, and network communication to support a shared controlled forensic relationship. Its contribution is evaluated against conventional pairwise graph reasoning.

The third contribution is a Generative Attack-Sequence Reconstruction Engine that estimates missing benchmark-defined stages using the evidence structure generated by NSFRH. GASRE penalizes candidates that lack sufficient forensic support, thereby distinguishing evidence-constrained reconstruction from unrestricted generation of a plausible attack narrative.

The fourth contribution is a Counterfactual Malware-Evolution Simulator that models prospective threat variations under specified interventions. CMES outputs are explicitly labelled as counterfactual scenarios and are excluded from observed-evidence counts, historical reconstruction ground truth, and primary classification evaluation.

The fifth contribution is a Cognitive Threat Attribution and Evidence Confidence Optimizer that performs confidence-aware attribution among benchmark-defined candidate threat categories. CTAECO integrates alignment confidence, symbolic-rule satisfaction, reconstructed-stage support, and predictive uncertainty and permits an inconclusive outcome when evidence confidence is insufficient.

The sixth contribution is a leakage-controlled and statistically transparent evaluation protocol. Duplicate-aware grouping prevents related records from crossing partitions, while temporal- and family-disjoint evaluations examine chronological drift and unseen-family generalization. Label-permutation and identifier-only controls assess leakage and metadata-based shortcut learning, and matched statistical analysis quantifies variability across repeated executions.

The seventh contribution is a module-level, alternative-design, and computational evaluation. The experimental analysis separately evaluates CTEEN, NSFRH, GASRE, CMES, and CTAECO; compares optimal transport with dynamic time warping, hypergraph reasoning with pairwise graph reasoning, evidence-constrained with unconstrained reconstruction, and calibrated with uncalibrated attribution; and reports model size, training time, peak memory, inference latency, and throughput. The complete framework introduces non-trivial computational overhead, and its deployment on resource-constrained platforms may therefore require model compression and inference optimization. (Reviewer 3, Comment 3) Computational profiling additionally quantifies model size, memory consumption, training time, inference latency, and throughput.

The methodological novelty does not arise from claiming that optimal transport, hypergraph learning, generative reconstruction, counterfactual simulation, or confidence calibration is independently new. Instead, it lies in their evidence-constrained and provenance-preserving interaction within a traceable multimodal forensic pipeline. Each module receives a defined output from the preceding stage and contributes to a separately evaluated forensic task.

The remainder of this paper is organized as follows. Section II critically reviews malware analysis, multimodal evidence fusion, forensic reasoning, generative reconstruction, counterfactual analysis, and confidence-calibration methods. Section III describes the proposed architecture and mathematical formulation. Section IV presents the datasets, controlled benchmark construction, baseline protocol, and evaluation methodology. Section V reports and discusses the experimental findings. Section VI concludes the study and outlines its limitations and future research directions.

Related work

The Research relevant to multimodal cyber forensics spans malware detection, heterogeneous evidence analysis, forensic reasoning, generative reconstruction, uncertainty estimation, and evidence governance. The literature is critically examined according to four capabilities required by the present study: the evidence modalities processed, the ability to align asynchronous artifacts, support for reconstructing an attack sequence, and treatment of attribution uncertainty. This capability-oriented organization avoids a paper-by-paper catalogue and distinguishes technically related cyber-forensic methods from studies that contribute only general forensic-intelligence principles.

Single-modality malware detection

A substantial portion of malware research formulates analysis as supervised classification using a single static or behavioural evidence source. Semantic code-representation methods improve robustness against identifier renaming and shallow obfuscation by learning structural dependencies from executable representations[3]. Android-oriented studies use permissions, APIs, application components, and inter-application communication to distinguish malicious applications or identify collusive behaviour[9,12,26,40]. Attention-based methods identify discriminative

regions in malware representations and provide feature-level saliency[11], whereas optimization-based approaches reduce feature dimensionality and computational cost[37,39].

Comparative studies and systematic reviews indicate that learned representations generally outperform manually designed features on standard malware corpora[8,13,1424,25,26,36,38]. Lightweight models further demonstrate that useful malware classification can be deployed on resource-constrained endpoints[27]. These methods provide effective malware detection within their intended settings, but their outputs generally terminate at an artifact-level label. They do not establish how audio, textual, executable, memory, and network evidence relate across time or recover attack stages that are absent from the observed records.

Feature-level explanation is also different from forensic reasoning. A saliency or importance map can identify which bytes, permissions, API calls, or behavioural features influenced a prediction, but it does not explain how multiple artifacts jointly support delivery, execution, persistence, lateral movement, or command-and-control activity. Single-modality approaches therefore support detection and local interpretation but provide limited assistance for reconstructing a complete multimodal evidentiary sequence.

Multimodal and language-model-based malware analysis

Multimodal malware analysis relaxes the assumption that one representation is sufficient. Static–dynamic fusion combines executable characteristics with API, permission, or system-call behaviour and demonstrates that these evidence types provide complementary information[30,53]. Multimodal Pre-trained Fusion[16] extends this direction using modality-specific pre-trained encoders and learned fusion across multiple application views. This method provides a particularly relevant baseline because it demonstrates the effectiveness of heterogeneous pre-trained representations for Android malware classification.

However, the modalities used by these systems generally originate from the same application and are treated as simultaneously available. Their integration is consequently sample-level rather than incident-level and temporal. Such fusion cannot directly establish correspondence among a phishing recording, a malicious message or script, subsequent executable activation, a volatile-memory observation, and a delayed command-and-control session recorded on different clocks.

LLM-MalDetect[44] serializes application behaviour into textual form and uses a large language model to produce a malware decision and a natural-language rationale. This work demonstrates the value of general semantic representations in malware analysis. Nevertheless, flattening heterogeneous observations into a textual sequence can remove explicit temporal and relational structure. Moreover, a generated rationale is not inherently constrained by a forensic evidence structure, and verbalized confidence is not equivalent to statistically calibrated uncertainty.

Federated language-model architectures provide privacy-preserving distributed reasoning without centralizing raw information[54]. This capability is relevant to future distributed forensic deployment, but such architectures primarily address privacy-preserving semantic or network reasoning. They do not explicitly distinguish observed artifacts, reconstructed historical stages, and prospective counterfactual projections within a unified forensic report.

Recent language-model and multi-agent security studies have further demonstrated the value of semantic reasoning, coordinated decision-making, and adaptive threat analysis. The Swarm OpenAI framework integrates large language models, transformers, and swarm intelligence for detecting dual-vector attacks in social networks[55]. RaSA-BoDX formulates cyberbullying detection and mitigation as a knowledge-constrained, multi-agent reasoning process rather than a conventional surface-level classification task[56]. Magteon-Turing L3TM similarly combines language models and ensemble mechanisms for adaptive social-network threat detection[57]. In network security, an intrusion detection and prevention framework has demonstrated the practical use of learned detection for brute-force and distributed denial-of-service attacks in cloud environments[58]. Recent critical analysis has also emphasized the need to move beyond classification-only security systems toward context-aware and knowledge-guided reasoning[59]. Although these studies do not provide direct malware-forensics baselines, they

support the use of semantic reasoning, multiple analytical agents, adaptive threat interpretation, and explainable decision mechanisms in contemporary cybersecurity systems.

Evidence-centric digital forensics

Evidence-centric studies begin with forensic artifacts rather than only with a classification objective. Family-specific investigations identify reproducible indicators from executable, device, or application evidence[1]. Memory-forensic methods demonstrate that volatile-memory residue can distinguish malware families and expose behavioural characteristics that may not be evident through static analysis[22]. Flow-based methods identify command-and-control activity without requiring payload inspection[28], whereas cross-domain localization techniques identify malicious subregions within binary and image-derived representations[51].

Autoencoder-based representations reduce dependence on extensive labelled corpora[52], while few-shot meta-learning supports adaptation to previously unseen malware families[54]. Blockchain-supported forensic systems additionally protect evidence integrity by maintaining tamper-evident acquisition and custody records[2,23]. These studies extend malware analysis beyond conventional closed-set classification, but the available evidence remains bounded by its original representation. A memory dump represents a particular observation point, a network flow cannot independently identify its responsible executable process, and a localized binary region does not establish its temporal relationship with delivery or communication evidence.

Consequently, investigators must still correlate the outputs of these systems manually. Existing evidence-centric approaches generally do not place heterogeneous artifacts on a shared temporal-semantic representation, model several related artifacts through a higher-order forensic structure, or reconstruct attack stages that were not directly captured.

Explainability, robustness, and confidence

Explainable malware detection commonly applies attention, feature importance, or SHAP-based attribution to identify the characteristics influencing a model decision[11,22,50]. These techniques improve sample-level transparency but do not provide a traceable relationship among artifacts collected from different evidence modalities.

Adversarial research further demonstrates that high benchmark accuracy can conceal sensitivity to deliberately constructed perturbations or distribution changes[49]. This finding is important in cyber forensics because a model should not retain high confidence when evidence lies outside the distribution represented during training. Existing adversarial studies, however, generally reveal detector vulnerability without providing an integrated mechanism for reducing confidence, reconstructing missing evidence, or representing alternative attack trajectories.

Few-shot attribution methods can adapt to unseen malware families using small support sets[54], but similarity and softmax scores do not necessarily represent calibrated probabilities. Likewise, natural-language confidence produced by an LLM does not provide a measurable error characteristic[44]. Confidence calibration and an explicit inconclusive outcome are therefore necessary when model outputs are intended to support investigative prioritization or candidate-category attribution.

Evidence integrity and forensic governance

Evidence-governance research emphasizes validity, transparency, reproducibility, human oversight, provenance preservation, and reportable error characteristics for AI-assisted forensic analysis[19,43]. Blockchain-based approaches strengthen evidence integrity and custody tracking[2,23], but generally treat the analytical output as an externally generated record. They protect the evidence and its audit trail without explaining how heterogeneous artifacts were temporally correlated or how a reconstructed attack-stage hypothesis was obtained.

The present work does not claim automatic legal or courtroom admissibility. It adopts the narrower technical requirements of provenance separation, traceable reasoning, calibrated uncertainty, and an inconclusive decision

when the available evidence is insufficient. Observed evidence, reconstructed historical-stage hypotheses, and prospective counterfactual scenarios are therefore treated as separate output categories.

Transferable cross-domain forensic-intelligence principles

Broader studies in questioned-document analysis, illicit-drug profiling, forensic medicine, odontology, microbiology, neuropathology, toxicology, psychiatry, psychology, phenotypic intelligence, contraband analysis, and forensic genetics emphasize cross-case linkage, contextual interpretation, evidential provenance, probabilistically qualified conclusions, and expert oversight[4–6-7,10,15,17,18,20,21,2931–3435,4145–47,48]. These studies are retained only as a consolidated source of transferable forensic-intelligence principles; they are not treated as technical precedents, experimental baselines, or evidence of multimodal cyber-forensic performance.

Critical comparison of closely related studies

Table 1 contains only the studies that are most directly related to the technical objectives of the proposed framework. The comparison focuses on the evidence processed, the validated methodological contribution, and the remaining limitation relevant to multimodal forensic reasoning.

Table 1 demonstrates that the closest studies individually address semantic malware representation, multimodal fusion, memory evidence, network behaviour, localization, unseen-family adaptation, explainability, robustness, or evidence governance. However, no reviewed method jointly performs temporal alignment of independently observed evidence, higher-order relational reasoning, evidence-constrained reconstruction, explicit separation of counterfactual projections, and calibrated candidate-category attribution within a traceable forensic pipeline. Non-cyber forensic studies are deliberately excluded from the comparison because they do not provide technically valid malware-analysis baselines.

Research gap and positioning of the proposed framework

Table 2 maps the unresolved literature gaps to the five modules of the proposed framework. The five modules are consequently not independent contributions assembled for coverage; each answers a specific and demonstrable deficiency in the corpus reviewed above, and together they close the loop from raw heterogeneous evidence to a defensible investigative narrative. CTEEN addresses temporal alignment, NSFRH addresses relational and rule-governed explanation, GASRE addresses constrained recovery of unobserved stages, CMES separates counterfactual projection from the evidential record, and CTAECO attaches calibrated confidence to attribution. To ensure that this claim is tested against the strongest available prior work rather than against weak baselines, the proposed framework is evaluated comparatively against Cyber Code Intelligence[3], AI-Based Malware Analysis[8], Multimodal Pre-trained Fusion[16] and LLM-MalDetect[44], which represent, respectively, the semantic single-modality, hybrid behavioural, multimodal pre-trained and large-language-model lines of research identified in this review.

Proposed integrated multimodal generative cyber forensics network

The proposed Integrated Multimodal Generative Cyber Forensics Network is designed as a sequential forensic-reasoning pipeline in which heterogeneous evidence is encoded, temporally aligned, semantically correlated, generatively reconstructed, counterfactually simulated, and finally transformed into confidence-aware forensic outputs. The framework consists of five connected modules: the Cross-Modal Temporal Evidence Encoding Network (CTEEN), Neuro-Symbolic Forensic Relationship Hypergraph (NSFRH), Generative Attack-Sequence Reconstruction Engine (GASRE), Counterfactual Malware-Evolution Simulator (CMES), and Cognitive Threat Attribution and Evidence Confidence Optimizer (CTAECO).

Table 1 Critical comparison of the most relevant prior studies.

Ref.	Evidence and primary task	Core mechanism and validated contribution	Remaining technical limitation
[3]	DEX bytecode, API graphs, and opcode sequences for Android malware classification	Semantic code representations improve resistance to identifier renaming and shallow obfuscation	Static, single-platform analysis without asynchronous evidence alignment, attack-stage reconstruction, or scalibrated attribution
[8]	Static, dynamic, and hybrid malware evidence for detection and mitigation	Organizes behavioural and learned malware-analysis mechanisms and provides a relevant AI-based comparison	Analysis generally terminates at detection or mitigation without a unified evidence timeline or reconstructed forensic sequence
[11]	Binary and image-derived malware representations	Attention identifies discriminative regions and provides inspectable sample-level saliency	Explanation remains within one artifact and does not represent cross-modal or stage-level forensic relationships
[16]	Multiple application views, including code, image-derived, and structural representations	Pre-trained modality-specific encoders demonstrate the value of heterogeneous feature fusion	Views originate from the same application and are fused without explicit temporal alignment; the output remains classification-oriented
[22]	Volatile-memory process, DLL, handle, and behavioural artifacts	Memory evidence supports malware-family discrimination and feature-level explanation	A memory dump represents one observation point and is not linked to preceding delivery, executable, or network events
[28]	Network-flow and beaconing characteristics for command-and-control detection	Detects command-and-control communication without dependence on payload content	Traffic-only evidence cannot independently identify the responsible process, binary, or delivery mechanism
[30]	Combined static and dynamic application evidence	Static–dynamic fusion validates the complementarity of executable and runtime characteristics	Sample-level concatenation does not align asynchronous sources or model sequential and higher-order forensic relationships
[44]	Textual serialization of Android-application behaviour	LLM-based semantic reasoning produces malware decisions and human-readable rationales	Evidence is flattened into text, the rationale is not constrained by a forensic graph, and confidence is not calibrated
[49]	Adversarial perturbations against learned IoT malware detectors	Demonstrates that high benchmark performance can conceal vulnerability to distribution manipulation	Identifies detector brittleness without providing evidence reconstruction, uncertainty calibration, or defensive reasoning
[51]	Binary and image-derived representations for malicious-region localization	Extends classification to subsample localization across representation domains	Localization remains within one artifact and provides no temporal ordering across evidence sources
[54]	Few-shot evidence episodes for unseen-family detection and attribution	Supports adaptation to previously unseen families using limited labelled support	Uses single-modality episodes; similarity-based attribution is not calibrated, and missing stages are not reconstructed
[19,43]	Governance and admissibility requirements for AI-supported forensic evidence	Establish the need for validity, transparency, oversight, and reportable error characteristics	Normative rather than computational; provide no mechanism for multimodal traceability, constrained reconstruction, or calibrated attribution

The methodological contribution does not depend on claiming that optimal transport, hypergraph learning, generative reconstruction, counterfactual simulation, or confidence calibration is independently new. Instead, it lies in the evidence-constrained interaction of these components. Each module receives a formally defined input from the preceding stage and produces an output that is independently evaluated in Section V. The framework also maintains an explicit distinction among observed evidence, reconstructed historical stages, and prospective counterfactual projections.

Table 2 Mapping from unresolved literature gaps to the five modules of the proposed framework.

Identified gap	Evidence in the reviewed literature	Investigative consequence	Proposed response in this work
G1. Temporal misalignment of asynchronous evidence	Multimodal and hybrid systems fuse views that originate from a single artefact and are treated as co-occurring[16,30] traffic[28], memory-[22]and endpoint[27]evidence are each modelled in isolation.	Events observed on different clocks cannot be ordered, so the investigator cannot state what happened before what.	CTEEN: timestamp-level cross-modal encoding that projects audio, script, binary, traffic and memory evidence into a shared temporal embedding space.
G2. Relational reasoning confined to pairwise or feature-level association	Explanation is delivered as per-feature attribution[2250], or as an unconstrained textual rationale[44]; relational forensic doctrine remains conceptual[4,7,46]	Explanations justify a score but do not state how artefacts relate causally, which weakens evidentiary value.	NSFRH: neuro-symbolic reasoning that applies explicit forensic rules over the learned evidence graph and returns a traceable inference chain.
G3. No constrained recovery of unobserved attack stages	Detection pipelines report on what was captured[3,8,11,37,40] missing links in intelligence graphs are left as gaps[6]; localisation is bounded by the artefact in hand[51].	Partial telemetry yields a partial account, and the investigator is left to infer the deleted or unlogged stages manually.	GASRE: generative reconstruction of the attack sequence under evidence-consistency constraints, so recovered stages remain compatible with observed artefacts.
G4. Retrospective analysis conflated with forward projection	Predictive treatments are either macroscopic and parameterised[42]or doctrinal[29,34] while detection studies remain closed-world[12,26] and adversarial evolution is studied only from the attacker side[49].	Systems cannot distinguish what the evidence shows from what the campaign is likely to do next, so projections contaminate findings of fact.	CMES: counterfactual malware evolution simulation kept architecturally separate from the evidential record while conditioned on it.
G5. Attribution reported without calibration	Attribution is emitted as an uncalibrated similarity or softmax score[54,51], or as verbalised model confidence[44]; governance[19]and admissibility[43]analyses require error rates that no reviewed system produces.	Uncalibrated confidence cannot support proportionate investigative or judicial decisions.	CTAECO: confidence-aware attribution with calibrated, reportable uncertainty in the likelihood-ratio reporting tradition of forensic science[32,47].

Multimodal forensic-evidence representation

Let the four forensic modalities be audio evidence A, textual evidence T, binary evidence B, and network-traffic evidence N. The complete multimodal evidence space is represented as

$$\mathcal{X} = \{A, T, B, N\} = \left\{ \{a_t\}_{t=1}^{T_A}, \{s_t\}_{t=1}^{T_T}, \{b_t\}_{t=1}^{T_B}, \{n_t\}_{t=1}^{T_N} \right\}. \tag{1}$$

As defined in (1), a_t represents acoustic and speech-semantic phishing features, s_t denotes malicious message, command, or script tokens, b_t denotes opcode or byte-level executable traces, and n_t represents network-flow vectors. The traffic representation may include packet length, inter-arrival time, protocol flag, port characteristics, source–destination transitions, and payload statistics.

The four evidence modalities may be drawn from independently collected public repositories. Therefore, shared class membership is not treated as evidence that two artifacts originated from the same real incident. Positive and negative cross-modal relationships are defined only by the controlled episode-construction, grouping, compatibility, and partitioning rules specified in Section IV.

Because the modalities differ in their raw structure, sampling frequency, and semantic granularity, each evidence stream is transformed using a modality-specific encoder:

$$z_m(t) = f_m(x_m(t); \theta_m), \quad m \in \{A, T, B, N\}, \tag{2}$$

where $x_m(t)$ is the raw observation from modality m, $f_m(\cdot)$ is its encoder, θ_m represents the trainable encoder parameters, and $\boldsymbol{z}_m(t)$ is the resulting continuous representation. As shown in (2), the audio encoder extracts spectral, prosodic, and speech-semantic features; the textual encoder captures syntactic and semantic malicious intent; the binary encoder models opcode transitions and byte distributions; and the network encoder represents packet-flow dynamics.

Abrupt adversarial changes are emphasized through a derivative-enhanced representation:

$$\boldsymbol{g}_m(t) = \frac{\partial \boldsymbol{z}_m(t)}{\partial t} + \lambda_m \frac{\partial^2 \boldsymbol{z}_m(t)}{\partial t^2}, \tag{3}$$

where λ_m controls the contribution of the second-order temporal variation. Equation (3) captures short-term and accelerated behavioural changes that may correspond to payload activation, command execution, sudden traffic bursts, or other malicious transitions. The representation is particularly relevant when forensic activity appears as a discontinuity rather than a smoothly evolving signal.

Figure 2 presents the complete processing sequence. The modality-specific evidence is first encoded independently and subsequently supplied to CTEEN. The aligned representation is transformed into a forensic hypergraph by NSFRH, which provides the evidence structure required by GASRE. CMES operates on the reconstructed representation without modifying the observed historical evidence, while CTAECO produces confidence-aware classification, candidate-category attribution, and forensic-reliability outputs.

Cross-modal temporal evidence encoding network

Phishing instructions, malicious scripts, executable activation, and network communication may occur at different moments and sampling resolutions. Consequently, exact timestamp matching can miss valid forensic relationships, whereas a wide fixed temporal window can introduce unrelated associations. CTEEN addresses this problem through optimal-transport-based temporal-semantic synchronization.

Figure 2. Technical architecture of the proposed Integrated Multimodal Generative Cyber Forensics Network, showing modality-specific evidence encoding, temporal-semantic alignment through CTEEN, higher-order evidence reasoning through NSFRH, evidence-constrained attack-stage reconstruction through GASRE, separately identified counterfactual threat simulation through CMES, and confidence-calibrated candidate-category attribution through CTAECO.

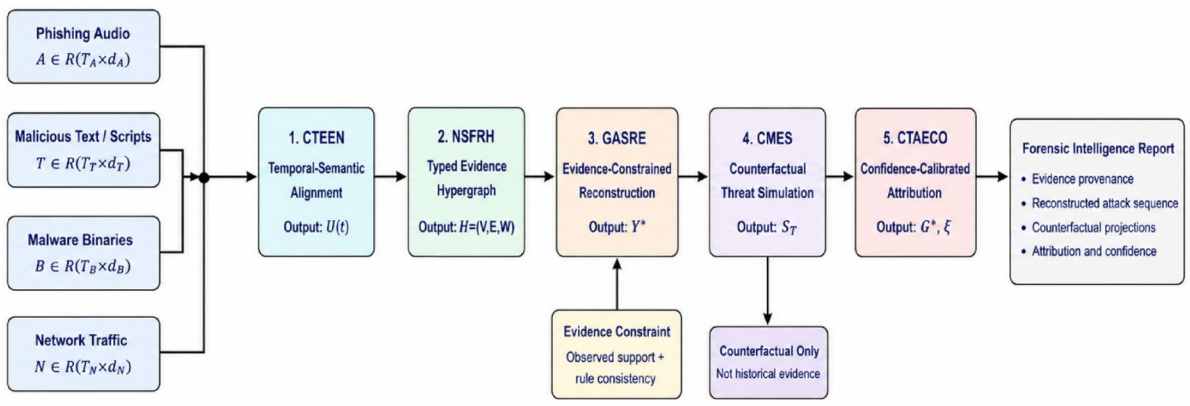


Fig. 2 Technical architecture of the proposed Integrated Multimodal Generative Cyber Forensics Network, showing modality-specific evidence encoding, temporal-semantic alignment through CTEEN, higher-order evidence reasoning through NSFRH, evidence-constrained attack-stage reconstruction through GASRE, separately identified counterfactual threat simulation through CMES, and confidence-calibrated candidate-category attribution through CTAECO.

For modalities m and k, the optimal coupling is estimated as

$$\pi^{*}_{m,k} = \underset{\pi \in \Pi(P_m, P_k)}{\arg\min} \int_{\mathcal{T}_m} \int_{\mathcal{T}_k} c_{m,k}(t,u) d\pi(t,u) \tag{4}$$

where $\Pi(P_m, P_k)$ is the set of admissible joint distributions with marginal temporal distributions P_m and P_k, and $c_{m,k}(t,u)$ is the temporal-semantic cost between evidence events observed at times t and u. As expressed in (4), temporal proximity alone is insufficient to create a positive cross-modal relationship because the coupling also considers semantic compatibility.

The aligned representation between modalities m and k is computed as

$$\boldsymbol{Z}_{m,k} = \int_{\mathcal{T}_m} \int_{\mathcal{T}_k} \pi^{*}_{m,k}(t,u)\, \psi\left(\boldsymbol{g}_m(t), \boldsymbol{g}_k(u)\right) dt\, du, \tag{5}$$

where $\psi(\cdot)$ is the cross-modal interaction function. Equation (5) allows a suspicious spoken instruction, a related script command, a compatible opcode transition, and a subsequent network-flow pattern to be correlated according to learned temporal-semantic compatibility rather than exact timestamp equality.

The complete Unified Temporal Evidence Matrix is defined as

$$\boldsymbol{U} = \sum_{m \in \mathcal{M}} \alpha_m \boldsymbol{Z}_m + \sum_{\substack{m,k \in \mathcal{M} \\ m<k}} \beta_{m,k} \boldsymbol{Z}_{m,k}, \quad \mathcal{M} = \{A, T, B, N\}, \tag{6}$$

where α_m is the reliability weight of modality m, $\beta_{m,k}$ is the dependency strength between modalities m and k, and $\boldsymbol{Z}_m$ is the unimodal representation. Equation (6) permits reliable evidence to contribute more strongly while reducing the effect of weak or inconsistent modalities.

The CTEEN alignment objective is formulated as

$$\mathcal{L}_{align} = \sum_{\substack{m,k \in \mathcal{M} \\ m<k}} \left\| Z_{m,k} - Z^{*}_{m,k} \right\|_F^2 + \eta \sum_t \left\| U_t - U_{t-1} \right\|_2^2, \tag{7}$$

where $\boldsymbol{Z}^{*}_{m,k}$ denotes the controlled correspondence target, $\|\cdot\|_F$ is the Frobenius norm, and η controls temporal regularization. The first term in (7) reduces cross-modal correspondence error, while the second limits unstable temporal fluctuations.

CTEEN therefore differs from direct multimodal concatenation because it generates a temporally and semantically weighted evidence surface before higher-order reasoning. It also differs from dynamic time warping because optimal transport supports soft and potentially many-to-many correspondence when a modality contains delayed, repeated, or missing events. This design choice is evaluated in Section V through a matched optimal-transport-versus-dynamic-time-warping experiment.

Neuro-symbolic forensic relationship hypergraph

The Unified Temporal Evidence Matrix generated by CTEEN is supplied to NSFRH. CTEEN determines when and to what extent evidence entities may correspond, whereas NSFRH determines how multiple aligned entities jointly support a forensic relationship. The forensic hypergraph is represented as

$$\mathcal{H} = (\mathcal{V}, \mathcal{E}, \boldsymbol{W}_E), \tag{8}$$

where $\mathcal{V}$ is the evidence-vertex set, $\mathcal{E}$ is the hyperedge set, and $\boldsymbol{W}_E$ contains the hyperedge confidence weights. As defined in (8), the vertices may represent voice intents, script functions, opcode blocks, executable behaviours, IP-flow patterns, file hashes, registry changes, or command-and-control traces.

Unlike a pairwise graph, one hyperedge can connect several evidence entities involved in the same controlled forensic relation. For example, a delivery-to-execution relation may jointly involve a phishing instruction, script command, executable trace, and outbound connection.

The vertex–hyperedge incidence matrix is defined as

$$\boldsymbol{H}(v,e)=\begin{cases}1, & v\in e,\\ 0, & v\notin e,\end{cases}\quad v\in\mathcal{V},\ e\in\mathcal{E}. \tag{9}$$

Equation (9) records the membership of each evidence entity in each forensic hyperedge. Hyperedges are created only for relationships that satisfy the controlled temporal-semantic and forensic-rule conditions. Similar class labels or repository membership alone do not establish a hyperedge.

The layer-wise hypergraph convolution is expressed as

$$\boldsymbol{R}^{(l+1)}=\sigma\left(\boldsymbol{D}_V^{-\frac{1}{2}}\boldsymbol{H}\boldsymbol{W}_E\boldsymbol{D}_E^{-1}\boldsymbol{H}^{\top}\boldsymbol{D}_V^{-\frac{1}{2}}\boldsymbol{R}^{(l)}\boldsymbol{\Theta}^{(l)}\right), \tag{10}$$

where $\boldsymbol{R}^{(l)}$ is the node representation at layer l, $\boldsymbol{D}_V$ and $\boldsymbol{D}_E$ are the vertex- and hyperedge-degree matrices, $\boldsymbol{\Theta}^{(l)}$ is the trainable projection matrix, and $\sigma(\cdot)$ is a nonlinear activation function. Through (10), evidence entities participating in a shared forensic relation can exchange information within a unified propagation step.

Symbolic cybersecurity knowledge is incorporated using the differentiable constraint

$$\mathcal{L}_{\text{rule}}=\sum\nolimits_{r\in\mathcal{R}}\max\left[0,\gamma_r-\rho_r(\mathcal{H},\boldsymbol{U})\right], \tag{11}$$

where $\mathcal{R}$ is the forensic-rule set, $\rho_r(\mathcal{H},\boldsymbol{U})$ is the satisfaction score of rule r, and γ_r is its minimum confidence threshold. Equation (11) penalizes relations that contradict the defined forensic-stage dependencies. These rules constrain the representation but do not independently prove that an event occurred.

The complete NSFRH objective is

$$\mathcal{L}_{NSFRH}=\mathcal{L}_{hyp}+\lambda_r\mathcal{L}_{rule}+\lambda_s\sum\nolimits_{e\in\mathrm{E}}w_e\sum\nolimits_{i,j\in e}\left\|R_i-R_j\right\|_2^2, \tag{12}$$

where $\mathcal{L}_{\text{hyp}}$ denotes the hypergraph task loss, λ_r and λ_s are regularization coefficients, and w_e is the confidence assigned to hyperedge e. The final term of (12) promotes semantic consistency among evidence entities connected by high-confidence hyperedges.

NSFRH complements CTEEN rather than duplicating it: CTEEN estimates controlled temporal-semantic correspondence, whereas NSFRH represents the higher-order forensic relationships supported by that correspondence. The necessity of this formulation is evaluated in Section V by replacing the hypergraph with a pairwise graph under an otherwise identical protocol.

Generative attack-sequence reconstruction engine

The forensic knowledge representation generated by NSFRH is supplied to GASRE. Let the attack sequence be $Y=\{y_1,y_2,\ldots,y_K\}$, where each y_k denotes a benchmark-defined forensic stage, such as initial lure, payload delivery, execution, privilege escalation, persistence, lateral movement, exfiltration, or command-and-control activity.

The conditional generation probability is formulated as

$$P_{\theta_G}(Y\mid\mathcal{H},\boldsymbol{U})=\prod\nolimits_{k=1}^{K}P_{\theta_G}(y_k\mid y_{<k},\mathcal{H},\boldsymbol{U}), \tag{13}$$

where θ_G represents the GASRE parameters and $y_{<k}$ denotes the preceding attack stages. Equation (13) ensures that each reconstructed stage is conditioned on both the preceding sequence and the NSFRH evidence structure.

Missing stages are recovered using the evidence-consistency objective

$$\hat{Y} = \arg\max_{Y \in y} \left[\log P_{\theta_G}(Y \mid \mathcal{H}, U) - \delta \sum_{k=1}^{K} \left\| \phi(y_k) - r_{\mathcal{H},k} \right\|_2^2 \right] \tag{14}$$

where $\phi(y_k)$ maps candidate stage y_k into the hypergraph embedding space, $\boldsymbol{r}_{\mathcal{H},k}$ is the evidence representation supporting stage k, and δ controls faithfulness to observed evidence. The second term in (14) penalizes generated stages that are plausible at the language level but insufficiently supported by the forensic hypergraph.

The reconstruction-confidence score is calculated as

$$C_{\text{rec}} = 1 - \frac{-\sum_{k=1}^{K} p_k \log(p_k + \epsilon)}{\log K}, \tag{15}$$

where p_k is the normalized probability associated with reconstructed stage k, and ϵ prevents numerical instability. As indicated by (15), C_{rec} approaches one when the reconstructed sequence has low normalized entropy and high confidence.

GASRE addresses missing information caused by deleted logs, partial packet captures, incomplete recordings, or unavailable intermediate artifacts. Nevertheless, its output is limited to reconstructing controlled benchmark stages supported by the available evidence. It is not presented as unrestricted generation of a complete real-world attack narrative. Section V evaluates the role of the evidence constraint by comparing GASRE with an unconstrained reconstruction alternative.

As shown in Fig. 3, the framework maintains three distinct information states. The first contains observed multimodal evidence. The second contains stages reconstructed through the evidence-consistency condition in (14). The third contains prospective counterfactual trajectories generated by CMES. Counterfactual trajectories are neither returned to the observed-evidence set nor treated as historical ground truth.

Counterfactual malware-evolution simulator

CMES evaluates how the reconstructed malware episode could evolve under specified adversarial interventions. Its outputs represent prospective analytical scenarios rather than events established by historical evidence. A counterfactual attack state is defined as

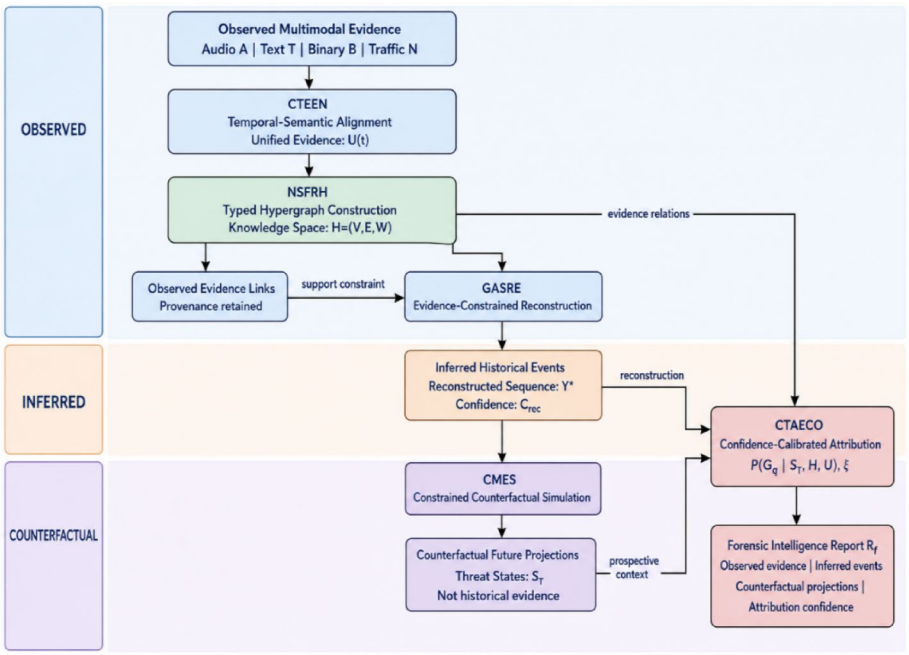


Fig. 3 End-to-end procedural flow of the proposed framework, distinguishing observed multimodal evidence, controlled temporal-semantic correspondence, higher-order forensic relationships, evidence-constrained historical-stage reconstruction, separately labelled counterfactual projections, and confidence-calibrated candidate-category attribution.

$$\tilde{\boldsymbol{y}}_k^{(c)} = \boldsymbol{y}_k + \Delta_k^{(c)}, \tag{16}$$

Equation (16) represents the intervention-induced modification of the reconstructed malware state while preserving its separation from historical evidence. Where $\boldsymbol{y}_k$ is the original stage representation and $\Delta_k^{(c)}$ is the learned mutation vector associated with counterfactual condition c. The intervention may represent a modified payload, altered command server, changed obfuscation mechanism, increased packet delay, or polymorphic executable behaviour.

The latent malware-evolution trajectory is modelled through the stochastic differential Eq.

$$d\boldsymbol{S}_t = \mu\,(\boldsymbol{S}_t, t, c)\,dt + \sigma\,(\boldsymbol{S}_t, t, c)\,d\boldsymbol{W}_t, \tag{17}$$

where $\boldsymbol{S}_t$ is the latent malware-evolution state, $\mu(\cdot)$ represents deterministic adversarial progression, $\sigma(\cdot)$ represents mutation uncertainty, and $d\boldsymbol{W}_t$ is the Wiener noise process. Equation (17) models a distribution of possible variations instead of asserting one deterministic future event.

The probability of a future threat state is obtained as

$$P(\boldsymbol{S}_T \mid \boldsymbol{S}_0, c) = \int P(\boldsymbol{S}_T \mid \boldsymbol{S}_{0:T}, c)\, P(\boldsymbol{S}_{0:T} \mid \boldsymbol{S}_0, c)\, d\boldsymbol{S}_{0:T}. \tag{18}$$

Equation (18) estimates the conditional probability of a future state under the specified intervention and controlled modelling assumptions. It does not establish that the projected behaviour will occur during operational deployment.

Figure 3. End-to-end procedural flow of the proposed framework, distinguishing observed multimodal evidence, controlled temporal-semantic correspondence, higher-order forensic relationships, evidence-constrained historical-stage reconstruction, separately labelled counterfactual projections, and confidence-calibrated candidate-category attribution.

The counterfactual simulation objective is defined as

$$\mathcal{L}_{CMES} = \mathbb{E}\left[D\left(S_T, \tilde{S}_T^{(c)}\right)\right] + \lambda_\Delta \left\|\Delta^{(c)}\right\|_1 + \lambda_v \mathcal{L}_{valid}, \tag{19}$$

Equation (19) formalizes the gradient-level separation between observed evidence and counterfactual simulation. Here, $E_\theta(\cdot)$ is the observed-evidence encoder parameterized by θ, $\boldsymbol{x}$ denotes the observed multimodal evidence, and $\boldsymbol{z}_{\text{obs}}$ is its latent representation. The operator $\text{sg}(\cdot)$ applies stop-gradient to produce the immutable reference state $\tilde{z}_{obs}$. The counterfactual generator $G_\phi(\cdot)$, parameterized by ϕ, generates the intervention-conditioned latent state $z_{cf}^{(c)}$, where c specifies the intervention and $\in$ represents stochastic variation.

The divergence function $D(\cdot)$ encourages a behaviourally meaningful difference between the observed reference and its counterfactual realization. The L_1-regularization term, weighted by λ_Δ, restricts unnecessary latent perturbations, whereas $\mathcal{L}_{\text{valid}}$, weighted by λ_v, penalizes counterfactual transitions that violate the admissible forensic constraints. The final condition in (19) establishes that the CMES-specific loss has zero gradient with respect to the observed latent state. Consequently, CMES optimization updates the counterfactual branch without modifying the observed-evidence representation through the reference path.

Counterfactual states are maintained in a separate scenario buffer and are excluded from the observed-evidence set, episode labels, temporal-alignment ground truth, historical reconstruction targets, and primary classification targets. Thus, the proposed mechanism enforces separation at the latent-state, gradient, supervision, and evaluation levels. The counterfactual branch can use observed evidence as an immutable reference during simulation, but neither the simulated states nor their gradients can convert a prospective scenario into an observed or reconstructed forensic fact. (Reviewer 3, Comment 4)

Cognitive threat attribution and evidence confidence optimizer

CTAECO integrates temporal alignment, hypergraph reasoning, reconstructed stages, and counterfactual uncertainty to estimate attribution among the benchmark-defined candidate threat categories. Let G_q denote candidate category q, and let $\Omega = \{G_1, G_2, \ldots, G_Q\}$ represent the candidate set.

The candidate-category attribution probability is computed as

$$P\left(G_q \mid \boldsymbol{U}, \mathcal{H}, \hat{Y}, \boldsymbol{S}_T\right) = \frac{\exp\left(\zeta_q\right)}{\sum_{j=1}^{Q} \exp\left(\zeta_j\right)}, \tag{20}$$

Equation (20) converts the CTAECO candidate scores into normalized attribution probabilities over the benchmark-defined candidate categories. Where ζ_q is the internal attribution score for candidate G_q. The score is defined by

$$\zeta_q = \omega_1 \mathrm{sim}\left(\boldsymbol{S}_T, \Gamma_q\right) + \omega_2 \mathrm{sim}\left(\mathcal{H}, \Psi_q\right) + \omega_3 \mathrm{sim}\left(\boldsymbol{U}, \Lambda_q\right), \tag{21}$$

Equation (21) combines state similarity, hypergraph-behaviour consistency, and temporal-evidence compatibility when calculating each candidate score. Where Γ_q represents the candidate malware-evolution signature, Ψ_q denotes its hypergraph-behaviour signature, Λ_q represents its multimodal evidence pattern, and $\mathrm{sim}(\cdot)$ is a cosine- or kernel-based similarity function. The coefficients ω_1, ω_2, and ω_3 regulate their respective contributions.

The evidence-confidence score is estimated as

$$C_{\mathrm{evi}} = C_{\mathrm{align}} \times C_{\mathrm{rule}} \times \left(1 - \frac{H_{\mathrm{unc}}}{\log Q}\right), \tag{22}$$

where C_{align} is the temporal-alignment confidence, C_{rule} is the symbolic-rule satisfaction confidence, and H_{unc} is predictive entropy. Equation (22) decreases the final confidence when correspondence is weak, forensic-rule satisfaction is insufficient, or attribution uncertainty is high.

The CTAECO output is interpreted as attribution among the candidate categories represented in the controlled benchmark. It is not definitive attribution to a real individual, organization, or threat actor. When C_{evi} remains below the decision threshold defined in Section IV, the system returns an inconclusive result rather than forcing a candidate assignment. The calibration quality of CTAECO is evaluated in Section V using expected calibration error, Brier score, and a calibrated-versus-uncalibrated attribution experiment.

Unified optimization and final forensic output

The complete framework is optimized using the integrated objective

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{align} + \lambda_2 \mathcal{L}_{NSFRH} + \lambda_3 \mathcal{L}_{rec} + \lambda_4 \mathcal{L}_{CMES} + \lambda_5 \mathcal{L}_{attr} + \lambda_6 \|\Theta\|_2^2, \tag{23}$$

where $\mathcal{L}_{\mathrm{align}}$ is the CTEEN alignment loss, $\mathcal{L}_{\mathrm{NSFRH}}$ is the neuro-symbolic hypergraph loss, $\mathcal{L}_{\mathrm{rec}}$ is the GASRE reconstruction loss, $\mathcal{L}_{\mathrm{CMES}}$ is the counterfactual simulation loss, $\mathcal{L}_{\mathrm{attr}}$ is the candidate-category attribution loss, and $\|\Theta\|_2^2$ regularizes all trainable parameters. The coefficients λ_1 to λ_6 balance the six objective components.

During each CMES update, the observed reference state in the counterfactual term is detached from the computational graph. Therefore, the derivative of the CMES-specific loss with respect to the observed-reference path is zero, while the remaining observed-evidence objectives continue to optimize the evidential branch. This gradient boundary complements the data-level exclusion of counterfactual states from observed labels and reconstruction ground truth.

Although (23) provides a unified optimization objective, the model outputs remain task-specific. Classification, alignment, reconstruction, attribution, forecasting, and reliability have separate targets and evaluation

metrics. Their numerical values are therefore reported independently in Section V instead of being combined into a universal forensic-intelligence accuracy.

The final forensic output is represented as

$$R_f = \mathcal{F}\left(\boldsymbol{U}, \mathcal{H}, \hat{Y}, P(\boldsymbol{S}_T \mid \boldsymbol{S}_0, c), P(G_q \mid \boldsymbol{U}, \mathcal{H}, \hat{Y}, \boldsymbol{S}_T), C_{\text{evi}}\right), \tag{24}$$

where R_f is the final forensic-intelligence report, $\boldsymbol{U}$ contains controlled cross-modal evidence links, $\mathcal{H}$ contains higher-order forensic relationships, $\hat{Y}$ is the evidence-constrained reconstructed sequence, $P(\boldsymbol{S}_T \mid \boldsymbol{S}_0, c)$ contains separately labelled counterfactual projections, $P(G_q|\cdot)$ is the candidate-category attribution probability, and C_{evi} is the corresponding confidence.

As indicated by (24), the final report preserves the provenance of every output. Observed artifacts are identified as collected evidence, missing stages are labelled as reconstructed hypotheses, CMES outputs are marked as counterfactual projections, and the attribution result is accompanied by calibrated confidence. This separation supports forensic traceability and prevents simulated content from being presented as directly observed evidence.

Mechanistic contribution and experimental traceability

The five modules form a constrained reasoning chain:

Multimodal evidence → temporal - semantic alignment
→ higher - order forensic reasoning → evidence - constrained reconstruction
→ counterfactual simulation → confidence - aware candidate attribution.

CTEEN determines which asynchronous evidence entities can be meaningfully correlated. NSFRH determines how those entities jointly support a controlled forensic relationship. GASRE uses the hypergraph representation to reconstruct missing stages while penalizing unsupported candidates. CMES generates prospective variations without modifying the observed or reconstructed historical evidence. CTAECO propagates alignment confidence, symbolic-rule satisfaction, and predictive uncertainty into the final candidate-category attribution.

This interaction distinguishes the proposed framework from a collection of independent modules. Each mathematical design choice is connected to a corresponding experiment in Section V. The optimal-transport formulation in (4) is compared with dynamic time warping. The hypergraph formulation in (8)–(12) is compared with pairwise graph reasoning. The evidence-consistency formulation in (14) is compared with unconstrained generation. The confidence formulation in (22) is compared with uncalibrated attribution. Module-removal ablation additionally evaluates the distinct contribution of CTEEN, NSFRH, GASRE, CMES, and CTAECO.

The proposed methodology does not claim that the underlying mathematical operators are independently novel. Its contribution lies in their evidence-constrained, provenance-preserving, and confidence-aware integration for controlled multimodal cyber-forensic reasoning. Section IV describes the datasets, controlled episode-construction procedure, group-aware partitioning, baseline adaptation, temporal- and family-disjoint protocols, leakage audit, negative controls, statistical analysis, and computational environment used to evaluate these claims.

End-to-end forensic inference procedure

Algorithm 1 summarizes the complete forward procedure of the proposed framework. It connects the mathematical formulations of CTEEN, NSFRH, GASRE, CMES, and CTAECO without altering their individual objectives. The algorithm also preserves the distinction among directly observed evidence, reconstructed historical-stage hypotheses, and prospective counterfactual projections.

Algorithm 1 connects the complete forward pass of the proposed framework. Lines 3–10 implement modality-specific encoding and CTEEN alignment, Line 11 performs NSFRH reasoning, Lines 12–13 apply evidence-constrained GASRE reconstruction, Lines 14–17 generate separately labelled CMES projections, and Lines 18–23

perform calibrated CTAECO attribution with an inconclusive outcome. This procedure ensures that observed evidence, reconstructed historical stages, and prospective counterfactual trajectories remain explicitly separated.

Experimental methodology

This section describes the evaluated datasets, controlled multimodal benchmark construction, evidence preprocessing, data partitioning, baseline implementation, model configuration, evaluation measures, generalization tests, leakage controls, statistical analysis, ablation protocol, and computational-efficiency assessment. The experimental design incorporates controlled benchmark construction, group-aware partitioning, distribution-shift evaluation, leakage controls, statistical analysis, ablation studies, and computational-efficiency assessment.

Evaluated dataset variants

The proposed Multimodal Generative Cyber Forensics Network was evaluated using four source-specific dataset variants and one integrated controlled multimodal benchmark. The source-specific variants were VirusShare, CIC-MalMem-2022, the Android Malware Corpus, and the Malware Traffic Repository. The fifth variant was constructed from eligible audio, textual, binary, and network evidence obtained from the source collections described below. The complete source pool contained approximately 52,000 malware files, 1.8 million network-flow records, 24,000 malicious textual or script artifacts, and 12,500 phishing or social-engineering audio recordings. The represented threat categories included ransomware, banking Trojans, spyware, botnets, credential-oriented attacks, and command-and-control activity. VirusShare supplied executable binaries containing byte-, opcode-, and structure-oriented malware indicators. CIC-MalMem-2022 contributed memory and behavioural characteristics associated with malware execution. The Android Malware Corpus supplied Android permissions, APIs, application components, strings, textual indicators, and executable characteristics. The Malware Traffic Repository supplied malware-related packet-flow and command-and-control behaviour. The integrated controlled multimodal benchmark additionally incorporated eligible network evidence from CICIDS2017, textual evidence from publicly available phishing-email, phishing-URL, malicious-script, PowerShell, and exploit-text collections, and audio evidence from publicly available phishing-speech, scam-call, and social-engineering collections. These supporting collections were not treated as additional dataset-level evaluation variants. They provided modality-specific evidence for constructing the integrated benchmark.

The approximate values in Table 3 describe the source pools before duplicate removal, eligibility filtering, controlled episode construction, and group-aware partitioning. They do not represent the number of naturally paired multimodal incidents.

Dataset-specific classification was evaluated on VirusShare, CIC-MalMem-2022, the Android Malware Corpus, and the Malware Traffic Repository using the evidence representations available in each collection. These four variants should not be interpreted as complete four-modality datasets. Full audio–text–binary–traffic fusion was evaluated only on the Integrated Controlled Multimodal Benchmark. Consequently, CICIDS2017 and the supporting public textual and audio collections were not assigned separate dataset-level classification results. Their eligible records contributed only to the integrated multimodal evaluation. This distinction maintains consistency with the five dataset variants reported in Section V. For reproducibility, the integrated benchmark was generated deterministically after timestamp normalization, eligibility filtering, duplicate-group assignment, temporal compatibility checking, and episode-level grouping, using the configuration parameters and deterministic seeds specified in the experimental protocol. (Reviewer 3, Comment 2).

Table 3 Evaluated dataset variants and experimental roles.

Evaluated dataset variant	Primary evidence	Approximate source pool	Experimental role
Virus share	Executable binaries	Included within 52,000 malware files	Binary malware classification and executable-stage representation
CIC-MalMem-2022	Memory and behavioural evidence	Included within 52,000 malware records	Memory-based malware classification and behavioural analysis
Android malware corpus	Permissions, APIs, textual and executable indicators	Included within 52,000 malware records	Android malware classification and family-disjoint evaluation
Malware traffic repository	Malware-related network traffic	Included within 1.8 million flow records	Traffic classification and binary–traffic correspondence
Integrated controlled multimodal benchmark	Audio, text/scripts, binaries and network traffic	Constructed from eligible records in the source collections	Full multimodal alignment, reconstruction, candidate-category attribution and reliability evaluation

Controlled multimodal benchmark construction

The Integrated Controlled Multimodal Benchmark was constructed by grouping eligible evidence according to predefined threat-category, forensic-stage, temporal-window, and compatibility rules. A controlled episode could contain phishing or social-engineering audio, malicious textual or script evidence, executable evidence, and network activity associated with compatible benchmark categories and attack stages. The source repositories were collected independently and did not provide naturally paired four-modality incidents. Therefore, artifacts were not associated solely because they shared a malware label or broad threat category. A positive cross-modal correspondence required category compatibility, forensic-stage compatibility, compliance with the 60-s temporal-alignment window, and satisfaction of the source- and duplicate-group restrictions. Artifacts belonging to incompatible categories, unrelated forensic stages, or non-admissible temporal positions were treated as negative correspondences. Semantically similar evidence from incompatible episodes was included as hard negative evidence. This prevented the alignment task from being solved only through broad malware-category similarity.

To ensure deterministic construction of the Integrated Controlled Multimodal Benchmark, temporal compatibility was evaluated only after timestamp parsing, source-specific normalization, eligibility filtering, and duplicate-group assignment. Absolute timestamps originating from independently collected repositories were not interpreted as evidence that two artifacts belonged to the same operational incident. Instead, they were used only within the controlled episode-construction rules described in this section. Records containing valid timestamps were converted to a common time unit and evaluated using the inclusive 60-s admissibility window. Consequently, a candidate occurring exactly at the boundary was eligible, whereas a candidate outside the boundary was rejected. Malformed, irrecoverable, or missing timestamps were not imputed and such records were excluded from timestamp-derived positive pairs.

When several artifacts from the same modality satisfied the temporal condition, each candidate was independently tested for threat-category, forensic-stage, provenance, and duplicate-group compatibility. Temporal proximity alone was therefore insufficient to create a positive relationship. Candidates violating any mandatory compatibility condition were rejected, while semantically similar but incompatible artifacts could be retained as hard negatives when they belonged to distinct provenance and duplicate groups. Events sharing the same timestamp were retained as separate evidence entities and ordered using the predefined forensic-stage precedence rule. Exact and near-duplicate artifacts were assigned to a common evidence group before episode generation, preventing the same underlying artifact from producing multiple apparently independent correspondences. The construction scripts record the accepted evidence identifiers, rejected-candidate reason codes, normalized temporal values, episode identifier, configuration version, and deterministic seed for each generated episode. These

rules make the controlled benchmark reconstructible while preserving the distinction between benchmark-defined correspondence and naturally observed incident-level correspondence. (Reviewer 3, comment 2)

The controlled correspondence labels were used to evaluate CTEEN. Accordingly, cross-modal alignment accuracy measured the recovery of experimentally defined positive and negative relationships. It did not establish that independently collected artifacts originated from the same operational incident. The attack-stage sequences used to evaluate GASRE were derived from the controlled episode labels. Selected stages were masked during testing, and reconstruction performance measured whether GASRE recovered the withheld benchmark-defined stages in their permitted order. Reconstruction accuracy therefore measured evidence-constrained stage recovery and not unrestricted generation of a plausible attack narrative. The attribution task was similarly limited to the benchmark-defined candidate threat categories represented in the controlled corpus. It did not constitute definitive attribution to a real person, organization, or operational threat actor.

Evidence preprocessing

Audio recordings were resampled to 16 kHz and transformed into acoustic, prosodic, and speech-semantic representations. Amplitude normalization and silence handling were performed without using statistics from the test partitions. Textual evidence was tokenized using a maximum sequence length of 512 tokens. Longer emails, scripts, and textual artifacts were divided into semantically continuous segments. Shorter sequences were padded using masked positions that did not contribute to attention calculation. Executable evidence was transformed into opcode- or byte-level sequences having a maximum length of 1,024 elements. Corrupted files and records for which the required representation could not be extracted were excluded before partitioning. Network evidence was represented using 50 statistical traffic attributes. These included flow duration, packet counts, packet-length characteristics, inter-arrival times, protocol information, connection flags, port properties, source–destination transitions, payload statistics, and entropy-related features. All normalization parameters, vocabularies, preprocessing transformations, decision thresholds, and calibration parameters were estimated exclusively from the corresponding training or validation partition. No information derived from the held-out test labels was used during representation fitting, model selection, or calibration.

Group-aware partitioning

A record-wise random split was not used because exact duplicates, near-duplicate malware variants, related source records, or artifacts derived from the same controlled episode could otherwise appear in multiple partitions. Before splitting, each record was assigned to a group using exact content hashes, near-duplicate signatures, source identifiers, episode identifiers, and malware-family information when available.

All evidence associated with the same group was assigned exclusively to the training, validation, or test partition. The principal evaluation retained the base-paper division of 80% training, 10% validation, and 10% testing, but implemented it at the group level instead of the individual-record level. To estimate run-to-run variability, the group-aware division was repeated using five partition configurations and five deterministic training seeds. This produced 25 matched partition–seed observations for every evaluated method. Within each observation, all methods used identical training, validation, and testing groups. The validation partition was used for hyperparameter selection, early stopping, confidence calibration, and decision-threshold selection. The test partition remained unavailable until all model-development decisions were completed.

Baseline methods and fair comparison

The proposed framework was compared with Cyber Code Intelligence[3], AI-Based Malware Analysis[8], Multimodal Pre-trained Fusion[16], and LLM-MalDetect[44]. These corrected baseline names were used consistently throughout Sections IV and V. Cyber Code Intelligence[3] represented semantic code-oriented malware analysis.

AI-Based Malware Analysis[8] represented conventional artificial-intelligence-based malware feature processing. Multimodal Pre-trained Fusion[16] provided a recent comparison based on heterogeneous pre-trained representations. LLM-MalDetect[44] represented language-model-assisted semantic malware analysis. Performance values were not copied directly from the original baseline publications. Each method was implemented or adapted within the present experimental environment and evaluated using the same eligible evidence groups, preprocessing outputs, partitions, task definitions, and evaluation metrics. The adapted baselines received all supported modality representations through a common input interface. However, they were not provided with the proposed NSFRH hypergraph reasoning, evidence-constrained GASRE reconstruction, CMES counterfactual simulation, or CTAECO confidence optimizer. An equivalent validation-based hyperparameter-selection budget was used for every method.

The resulting comparisons represent the adapted baseline implementations under the present controlled protocol. Their lower performance should not be interpreted as evidence that the original methods are intrinsically inferior for the applications for which they were designed.

Model configuration and training procedure

The final configurations of the five proposed modules are summarized in Table 4.

Table 4 Model configuration and evaluation protocol.

Component	Final configuration	Component	Final configuration
Input evidence	Audio, textual/scripts, executable binaries and network traffic	GASRE reconstruction coefficient	0.40
Audio sampling rate	16 kHz	CMES latent evolution dimension	256
Maximum text length	512 tokens	CMES diffusion variance	0.15
Maximum opcode/byte sequence	1,024 elements	CTAECO attribution embedding	128 dimensions
Network representation	50 statistical traffic features	CTAECO confidence-calibration weight	0.25
CTEEN projected representation	512 dimensions	Optimizer	AdamW
CTEEN temporal-alignment window	60 s	Initial learning rate	1×10^{-4}
CTEEN alignment coefficient	0.05	Batch size	64
NSFRH reasoning layers	3	Maximum training epochs	200
NSFRH attention heads	16	Dropout rate	0.30
NSFRH node-embedding dimension	128	Primary partition	80:10:10 group-aware training, validation and testing
NSFRH symbolic-rule confidence threshold	0.75	Variability assessment	5 group-aware partitions × 5 deterministic seeds
GASRE transformer layers	6	Generalization evaluation	Temporally disjoint and family-disjoint testing
GASRE hidden dimension	768	Negative controls	Label-permutation and identifier-only classification
GASRE learning rate	1×10^{-4}	Statistical analysis	Repeated-measures ANOVA and Holm-corrected matched comparisons
		Computational measures	Parameters, training time, peak memory, latency and throughput

CTEEN used a 512-dimensional projected representation, a 60-s temporal-alignment window, a batch size of 64, and an alignment coefficient of 0.05. NSFRH contained three hypergraph-reasoning layers, 16 attention heads, 128-dimensional node embeddings, and a symbolic-rule confidence threshold of 0.75. GASRE contained six transformer layers with a hidden dimension of 768, a learning rate of 1×10^{-4}, and a reconstruction coefficient of 0.40. CMES used a 256-dimensional latent evolution representation and a diffusion variance of 0.15. CTAECO used a 128-dimensional attribution representation and a confidence-calibration weight of 0.25. The complete network was optimized using AdamW for a maximum of 200 epochs with an initial learning rate of 1×10^{-4}, batch size of 64, and dropout rate of 0.30. Training was stopped according to validation performance. Confidence calibration was performed after model fitting using validation predictions and was not adjusted using test results.

Evaluation tasks and metrics

Malware classification was evaluated using accuracy, precision, recall, F1-score, and area under the receiver-operating-characteristic curve. Dataset-specific classification results were reported for the five variants listed in Table 3. Cross-modal evidence alignment evaluated whether CTEEN recovered the controlled positive and negative correspondence labels. The evaluated settings were Voice–Text, Text–Binary, Binary–Traffic, Voice–Traffic, and Full Multimodal Fusion. The mean alignment accuracy reported in Section V was calculated across the five alignment settings. Therefore, the mean alignment value of 98.46% and the Full Multimodal Fusion result of 98.9% represent different performance summaries and are not inconsistent. Attack-sequence reconstruction accuracy measured agreement between the GASRE output and the withheld controlled episode stages. A reconstructed stage was counted as correct only when its stage category and permitted sequence position agreed with the benchmark target. Candidate-category attribution accuracy measured assignment among the benchmark-defined threat categories. Predictions that did not satisfy the CTAECO confidence requirement were reported as inconclusive. This task did not measure legal or definitive attribution to real threat actors. Forensic-intelligence reliability was evaluated through evidence consistency, narrative coherence, attribution confidence, threat-forecast reliability, and the predefined aggregate reliability score. These dimensions were reported separately because they measure different properties. The aggregate score was not described as universal forensic-intelligence accuracy. Confidence calibration was assessed using expected calibration error and Brier score. These measures were included because high attribution accuracy alone cannot establish that the corresponding confidence values are reliable.

Temporal- and family-disjoint evaluation

A temporally disjoint experiment was conducted to evaluate performance under chronological distribution shift. When reliable timestamps were available, chronologically earlier evidence groups were used for training and validation, while the latest non-overlapping groups were reserved for testing. Source-release or collection-order metadata was used when exact event timestamps were unavailable. No evidence group in the temporally held-out test set contributed to model fitting, hyperparameter selection, threshold selection, or confidence calibration. A family-disjoint experiment was conducted to evaluate previously unseen malware families. All evidence belonging to the evaluated target family, including its duplicate groups and derived controlled artifacts, was excluded from training, validation, model selection, and calibration. The family-disjoint result was reported separately from the primary group-aware result. Therefore, the standard controlled-split accuracy was not interpreted as unrestricted zero-day performance. Statements concerning previously unseen malware families were based only on the family-disjoint evaluation.

These evaluation settings were included to assess whether the proposed framework remains effective when the test evidence differs chronologically or by malware family from the evidence used during model development, thereby providing a more stringent assessment of generalization than a conventional random split. (Reviewer 3, Comment 1)

Leakage audit and negative controls

A partition-integrity audit was performed before model fitting. Exact binary and textual duplicates were identified through content hashes. Near-duplicate artifacts were grouped using similarity signatures. Source identifiers, episode identifiers, and malware-family assignments were examined across the partitions. When a conflict was identified, all associated records were assigned to the same partition. The audit reported the number of conflicts detected before group-aware partitioning and the number remaining afterwards. Zero remaining overlap was required for exact duplicates, near duplicates, repeated episode identifiers, repeated source identifiers, and the target family in family-disjoint testing. Two negative-control experiments were conducted. In the label-permutation experiment, training labels were randomly shuffled while the remaining training procedure was retained. In the identifier-only experiment, prediction was attempted using only repository, filename, source, collection, or episode identifiers. Performance near the balanced chance level in both controls was required to exclude obvious label leakage and metadata-based shortcut learning. Negative-control results were reported separately and were not included in the primary performance averages.

Ablation protocol

The component-ablation experiment evaluated the distinct contribution of CTEEN, NSFRH, GASRE, CMES, and CTAECO. One module was disabled at a time while the remaining architecture, preprocessing, data partitions, and training conditions were preserved. Four alternative-design comparisons were additionally conducted. Optimal-transport alignment was compared with dynamic time warping. NSFRH hypergraph reasoning was compared with pairwise graph reasoning. Evidence-constrained GASRE was compared with unconstrained reconstruction. Calibrated CTAECO was compared with an uncalibrated attribution configuration. These comparisons correspond directly to the mathematical design choices described in Section III. All ablation configurations used identical evidence groups, training conditions, and evaluation metrics.

Statistical analysis

Mean, standard deviation, and 95% confidence interval were calculated across the 25 matched partition–seed observations. Variation across different datasets, evidence pairings, attack categories, or reliability dimensions was not treated as run-to-run uncertainty. A repeated-measures ANOVA was used to evaluate the effect of the method on the matched performance observations. The statistical analysis reported the F-statistic, numerator and denominator degrees of freedom, probability value, and partial η^2. Proposed-versus-baseline comparisons were conducted using matched pairwise tests. Holm's sequential correction was applied to control the family-wise error rate. The pairwise results included the mean performance difference, 95% confidence interval, adjusted probability value, and standardized effect size. Statistical significance was accepted at a Holm-adjusted $p < 0.05$.

Computational-efficiency evaluation

The proposed framework and the adapted baselines were evaluated under the same computational environment. The reported measures included trainable parameter count, training time, peak memory consumption, inference latency per complete multimodal episode, and episode throughput. Training time covered the complete fitting procedure for one group-aware partition. Inference latency was measured after a fixed warm-up period using the same batch size, episode composition, and timing boundary for all methods. The computational cost of the proposed framework was interpreted in relation to its additional forensic operations, including temporal alignment, hypergraph reasoning, attack-stage reconstruction, counterfactual simulation, and confidence aggregation.

Controlled case illustration

The financial-sector scenario was used only as a controlled illustration of the report produced by the framework. It was not presented as deployment at an actual financial institution and was not treated as an independent held-out experiment. The controlled illustration contained 5,000 network sessions, 1,200 malware files, 850 phishing emails, and 300 voice-based social-engineering recordings, producing 7,350 evidence items. CTEEN aligned approximately 7,180 evidence items under the controlled episode rules, corresponding to approximately 97.7%.

GASRE outputs were labelled as reconstructed historical-stage hypotheses. CMES outputs were labelled as prospective counterfactual scenarios. CTAECO output was restricted to candidate categories represented in the controlled benchmark. The illustration was excluded from the primary averages, baseline comparisons, confidence intervals, and significance tests.

Scope of evaluation

The experimental protocol evaluates the internal effectiveness of the proposed framework under the defined controlled benchmark. Classification, alignment, reconstruction, attribution, forecasting, calibration, and reliability are separate tasks with different targets and error conditions.

The evaluation does not establish that independently collected artifacts originated from a common real incident. It also does not establish unrestricted zero-day detection, definitive real-world threat-actor attribution, courtroom admissibility, or universal operational reliability. The temporal- and family-disjoint experiments provide additional evidence of generalization, but their interpretation remains limited to the evaluated repositories and partition conditions.

Results and discussion

This section reports the malware-classification, controlled cross-modal alignment, attack-stage reconstruction, benchmark-defined candidate-category attribution, forensic-reliability, statistical, generalization, leakage-control, ablation, and computational-efficiency results. The evaluated baselines were Cyber Code Intelligence[3], AI-Based Malware Analysis[8], Multimodal Pre-trained Fusion[16], and LLM-MalDetect[44].

All methods were evaluated using the same eligible evidence groups, preprocessing outputs, group-aware partitions, validation procedure, and metric definitions described in Section IV. Performance values from the original baseline publications were not used as direct head-to-head results.

The results correspond to the controlled multimodal benchmark constructed from independently collected public repositories. They quantify performance under the specified episode-construction and evaluation protocol and should not be interpreted as universal performance on naturally co-collected operational incidents.

Malware-classification performance

Table 5 presents the average baseline performance across the five evaluated dataset variants and the dataset-specific performance of the proposed framework.

As reported in 5 A of Table 5, Cyber Code Intelligence[3] was the strongest baseline, achieving an average accuracy of 93.32%, an F1-score of 92.74%, and an AUC of 94.90%. Its comparatively strong result reflects the usefulness of semantic code representations. However, it does not explicitly model asynchronous evidence relationships, higher-order forensic associations, constrained attack-stage reconstruction, counterfactual projections, or confidence-aware attribution.

AI-Based Malware Analysis[8] achieved an average accuracy of 92.14%, whereas Multimodal Pre-trained Fusion[16] and LLM-MalDetect[44] achieved 90.36% and 91.34%, respectively. Multimodal Pre-trained Fusion[16]

Table 5 Overall and dataset-specific malware-classification performance (%) A: Average comparison across the five evaluated variants. B: Dataset-specific performance of the proposed framework.

Method	Accuracy	Precision	Recall	F1-score	AUC
Cyber code intelligence[3]	93.32	92.54	93.00	92.74	94.90
AI-based malware analysis[8]	92.14	91.16	91.72	91.42	93.68
Multimodal pre-trained fusion[16]	90.36	89.22	89.66	89.42	91.82
LLM-MalDetect[44]	91.34	90.22	90.54	90.36	93.04
Proposed framework	**99.26**	**99.16**	**99.30**	**99.20**	**99.66**
Dataset variant	Accuracy	Precision	Recall	F1-score	AUC
VirusShare	99.1	99.0	99.1	99.0	99.6
CIC-MalMem-2022	99.3	99.2	99.4	99.3	99.7
Android malware corpus	99.5	99.4	99.6	99.5	99.8
Malware traffic repository	99.0	98.9	99.0	98.9	99.5
Integrated controlled multimodal benchmark	99.4	99.3	99.4	99.3	99.7

and LLM-MalDetect[44] provide relevant modern comparisons, but their adapted implementations remain primarily classification- or representation-oriented. Their lower results under the present protocol should not be interpreted as evidence that the original methods are intrinsically inferior in their intended application domains.

The proposed framework achieved an average accuracy of 99.26%, precision of 99.16%, recall of 99.30%, F1-score of 99.20%, and AUC of 99.66%. The improvement is associated with the combined use of temporally aligned evidence, higher-order hypergraph relationships, and confidence-aware evidence aggregation. The contribution of individual modules and mathematical design choices is examined in Section V-F.

5B of Table 5 shows that the proposed framework did not produce an identical result on every dataset. Accuracy ranged from 99.0% to 99.5%, while F1-score ranged from 98.9% to 99.5%. The Android Malware Corpus produced the highest performance because its permissions, API calls, application components, strings, and executable indicators provide comparatively structured and discriminative information.

The Malware Traffic Repository produced the lowest accuracy and F1-score. Malware-related network flows may contain short-lived, overlapping, or encrypted behavioural patterns, making individual traffic records more difficult to distinguish. The Integrated Controlled Multimodal Benchmark achieved 99.4% accuracy and 99.3% F1-score, indicating that the eligible audio, textual, binary, and traffic representations remained complementary under the controlled episode protocol.

The four source-specific variants were evaluated using their available evidence representations. They should not be interpreted as complete four-modality datasets. Full audio–text–binary–traffic fusion was evaluated only through the Integrated Controlled Multimodal Benchmark.

Figure 4 visualizes the average values reported in 5 A of Table 5. The proposed framework maintained the highest result across all five classification measures. Its narrow metric range indicates balanced precision and recall under the controlled benchmark. However, it does not establish unrestricted performance under unseen operational distributions.

Cross-modal alignment and attack reconstruction

Table 6 reports controlled cross-modal evidence-alignment and attack-stage reconstruction performance. The two tasks are presented together because CTEEN supplies the aligned evidence representation used by NSFRH and GASRE, but they retain different targets and evaluation conditions.

Figure 4. Average accuracy, precision, recall, F1-score, and AUC of the proposed framework and four cyber- and malware-related baselines across the five evaluated dataset variants.

As shown in 6 A of Table 6, Text–Binary alignment produced the strongest pairwise result of 98.7%. Textual commands and scripts frequently contain semantic indicators corresponding with executable instructions or behaviours. Voice–Traffic alignment produced the lowest result of 98.0% because social-engineering intent

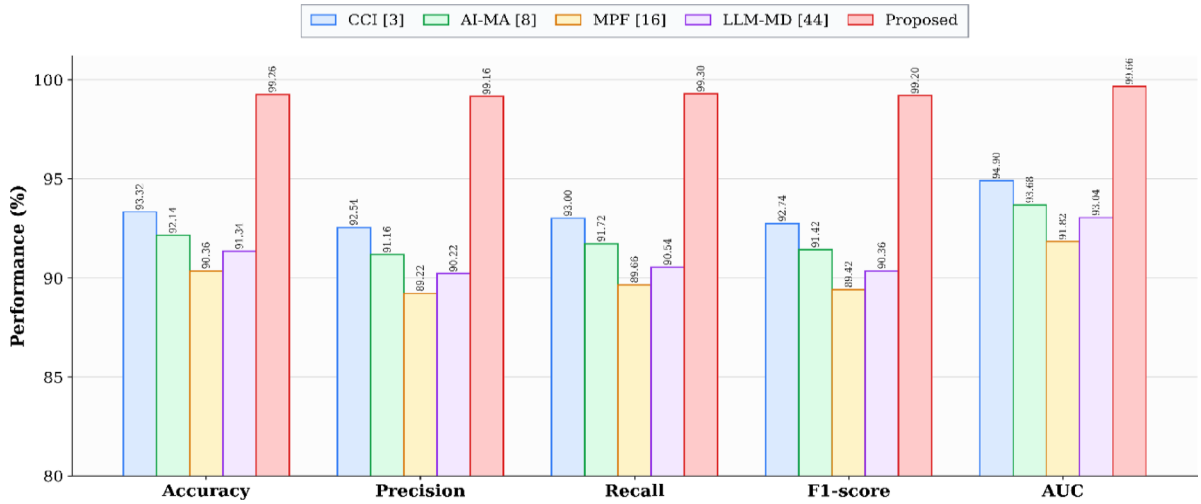


Fig. 4 Average accuracy, precision, recall, F1-score, and AUC of the proposed framework and four cyber- and malware-related baselines across the five evaluated dataset variants.

Table 6 Cross-modal alignment and attack-stage reconstruction accuracy (%) A: Cross-modal evidence alignment. B: Attack-stage reconstruction.

A. Evidence setting	Cyber code intelligence[3]	AI-based malware analysis[8]	Multimodal pre-trained fusion[16]	LLM-MalDetect[44]	Proposed framework
Voice–text	87.8	84.2	80.4	82.6	**98.4**
Text–binary	89.2	85.8	82.3	84.1	**98.7**
Binary–traffic	88.5	85.1	81.8	83.5	**98.3**
Voice–traffic	86.9	83.7	79.5	81.7	**98.0**
Full multimodal fusion	89.4	86.1	82.8	84.8	**98.9**
Mean	**88.36**	**84.98**	**81.36**	**83.34**	**98.46**
B. Attack category	Cyber code intelligence[3]	AI-based malware analysis[8]	Multimodal pre-trained fusion[16]	LLM-MalDetect[44]	Proposed framework
Ransomware	88.4	85.7	81.9	83.8	**98.5**
Banking Trojan	89.2	86.6	82.7	84.5	**98.9**
Spyware	87.6	84.9	80.8	82.4	**98.3**
Botnet	88.8	85.9	81.6	83.6	**98.6**
Multi-stage attack	90.0	86.8	82.9	84.8	**99.1**
Mean	**88.80**	**85.98**	**81.98**	**83.82**	**98.68**

and network communication are separated by more intermediate stages and therefore exhibit weaker direct correspondence.

Full Multimodal Fusion achieved 98.9% alignment accuracy. The mean across Voice–Text, Text–Binary, Binary–Traffic, Voice–Traffic, and Full Multimodal Fusion was 98.46%. The 98.9% and 98.46% values therefore describe full-fusion and overall mean performance, respectively, and are not inconsistent.

These alignment values measure recovery of the positive and negative correspondence labels created under the controlled episode-construction rules. They do not demonstrate that artifacts drawn from independent repositories originated from a common real incident.

6B shows that the proposed framework achieved its strongest reconstruction performance of 99.1% for controlled multi-stage attacks and its lowest result of 98.3% for spyware episodes. Multi-stage attacks provided more observable transitions across delivery, execution, persistence, and communication stages. Spyware episodes contained fewer externally observable events, which reduced the available support for recovering masked stages.

GASRE differs from unrestricted sequence generation because its candidate stages are evaluated against the NSFRH evidence structure. A stage unsupported by the available forensic relationships receives an evidence-consistency penalty. Reconstruction accuracy therefore measures recovery of the predefined controlled stages and not generation of a linguistically plausible narrative.

Figure 5 visualizes the task-level results in Table 6. Alignment varied from 98.0% to 98.9%, whereas reconstruction varied from 98.3% to 99.1%. The task-dependent variation demonstrates that alignment and reconstruction are separate prediction problems and should not be represented through a single common intelligence-accuracy value.

Candidate-category attribution and forensic reliability

Table 7 reports attribution among the candidate threat categories represented in the controlled benchmark and the corresponding forensic-reliability measures.

As reported in 7 A of Table 7, attribution performance ranged from 98.8% for botnet-operator categories to 99.3% for advanced persistent-threat categories. The comparatively stronger result for advanced persistent-threat categories was associated with longer controlled evidence chains and more distinctive combinations of tactics, techniques, and procedures. Botnet categories exhibited more overlapping traffic and executable behaviours, reducing their separability.

CTAECO combined the reconstructed stages, hypergraph relationships, temporal evidence representation, and confidence estimates rather than relying only on a malware-family prediction. Nevertheless, these values represent assignment among the candidate categories available in the controlled benchmark. They do not establish legal or definitive attribution to real organizations, operators, or individuals.

7B shows that evidence consistency produced the highest reliability value of 99.2%, whereas threat-forecast reliability produced the lowest value of 98.5%. This difference is expected because evidence consistency is evaluated against observed or predefined episode relationships, whereas forecasting contains additional counterfactual uncertainty.

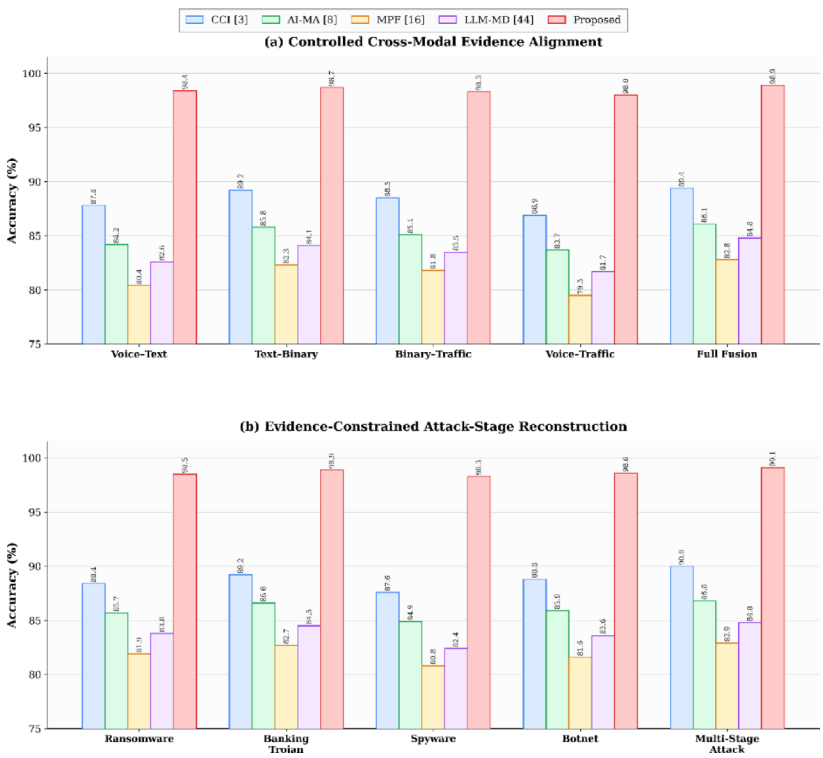


Fig. 5 (**a**) Cross-modal alignment and (**b**) evidence-constrained attack-stage reconstruction performance across evidence pairings and controlled attack categories.

Table 7 Candidate-category attribution and forensic-reliability performance (%) A: Benchmark-defined candidate-category attribution. B: Forensic reliability and confidence.

Candidate threat category	Cyber code intelligence[3]	AI-Based malware analysis[8]	Multimodal pre-trained fusion[16]	LLM-MalDetect[44]	Proposed framework
Ransomware groups	90.3	87.2	84.1	85.7	**99.0**
Banking malware groups	91.1	88.0	84.8	86.5	**99.2**
Botnet operators	89.5	86.4	83.5	85.1	**98.8**
Spyware campaigns	89.8	86.8	83.9	85.4	**98.9**
Advanced persistent-threat categories	91.4	88.3	85.2	86.9	**99.3**
Mean	**90.42**	**87.34**	**84.30**	**85.92**	**99.04**
Reliability measure	Cyber code intelligence[3]	AI-based malware analysis[8]	Multimodal pre-trained fusion[16]	LLM-MalDetect[44]	Proposed framework
Evidence consistency	91.6	89.2	87.4	88.8	**99.2**
Narrative coherence	90.8	88.4	86.1	87.5	**99.0**
Attribution confidence	90.5	87.9	85.7	86.8	**99.1**
Threat-forecast reliability	88.7	85.4	82.3	84.2	**98.5**
Overall intelligence reliability	90.4	87.7	85.4	86.8	**99.0**

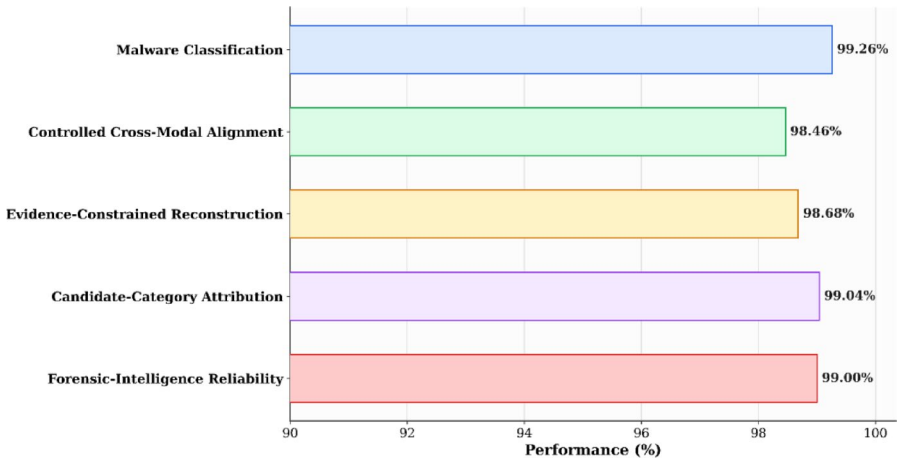


Fig. 6 Task-specific performance of the proposed framework for malware classification, controlled cross-modal alignment, evidence-constrained reconstruction, benchmark-defined candidate-category attribution, and forensic-intelligence reliability.

CMES outputs were retained as prospective analytical scenarios and were not presented as recovered historical evidence. Counterfactual predictions were also excluded from the primary classification and attack-reconstruction ground truth. The 99.0% overall reliability value represents the predefined aggregate of evidence consistency, coherence, attribution confidence, and forecast reliability. It should not be interpreted as universal forensic-intelligence accuracy. The separation was enforced at three levels: counterfactual states were stored separately from the evidential record, excluded from all observed and reconstruction targets, and generated from a stop-gradient copy of the observed latent reference. Thus, the counterfactual branch could be evaluated as a prospective analytical component without allowing simulated states or their supervision signals to become pseudo-observations. (Reviewer 3, Comment 4)

Figure 6 presents the five forensic tasks separately. Average classification accuracy was 99.26%, mean alignment accuracy was 98.46%, mean reconstruction accuracy was 98.68%, mean candidate-category attribution accuracy was 99.04%, and overall reliability was 99.0%. The figure prevents these distinct values from being reduced to a single undifferentiated intelligence-performance claim.

Run-to-run variability and statistical significance

Table 8 reports the mean, standard deviation, 95% confidence interval, and matched statistical comparison obtained from the 25 partition–seed observations.

As shown in 8 A of Table 8, the proposed framework achieved an accuracy of 99.26 ± 0.18%, an F1-score of 99.20 ± 0.21%, and an AUC of 99.66 ± 0.12%. Its accuracy confidence interval was 99.19–99.33%. These values quantify variability across the matched partition–seed observations and not variation among unrelated datasets or tasks.

The repeated-measures ANOVA identified a significant effect of method on the matched F1-score observations, $F(4, 96) = 1386.42$, $p < 0.001$, with partial $\eta^2 = 0.983$. Holm-corrected comparisons showed significant proposed-versus-baseline differences in every case. The smallest mean improvement was 6.46% points over Cyber Code Intelligence[3], while the largest was 9.78% points over Multimodal Pre-trained Fusion[16].

The statistical results provide run-to-run uncertainty, complete omnibus statistics, corrected pairwise tests, and standardized effect sizes. Their interpretation remains restricted to the matched controlled-benchmark observations.

Generalization, leakage audit, and negative controls

Table 9 reports performance under controlled, temporally disjoint, and family-disjoint testing, together with the partition-integrity audit and negative-control results.

As shown in Table 9A, the temporally disjoint test achieved 95.18% accuracy and a 94.71% F1-score, corresponding to reductions of 4.08 and 4.49% points, respectively, relative to the group-aware controlled split. The family-disjoint test produced a further reduction to 92.34% accuracy and a 91.37% F1-score. These declines demonstrate that the controlled group-aware results should not be interpreted as unrestricted real-world performance. Instead, the two disjoint protocols indicate that the framework retains useful forensic representations under chronological and malware-family distribution shifts while remaining sensitive to those shifts. Together with the leakage-controlled partitioning, these evaluations provide a more stringent assessment of generalization than sample-level random partitioning alone. (Reviewer 3, Comment 1)

The family-disjoint experiment was more challenging and produced 92.34% accuracy, 91.37% F1-score, and 94.65% AUC. The 6.92-percentage-point reduction in accuracy shows that the controlled-split result cannot be interpreted as unrestricted unseen-family performance. Nevertheless, the family-disjoint result indicates that the multimodal and higher-order representations retained useful information when the evaluated family was excluded from model development.

9B shows that record-wise partitioning would have introduced 1,284 exact binary conflicts, 736 near-duplicate evidence-group conflicts, 219 repeated episode identifiers, and 148 repeated source identifiers. Group-aware

Table 8 Fold–seed variability and matched statistical comparison A: Variability and 95% confidence intervals. B: Holm-corrected proposed-versus-baseline comparisons.

A. Method	Accuracy, mean ± SD (%)	F1-score, mean ± SD (%)	AUC, mean ± SD (%)	Accuracy 95% CI (%)
Cyber code intelligence[3]	93.32 ± 0.48	92.74 ± 0.53	94.90 ± 0.42	93.12–93.52
AI-based malware analysis[8]	92.14 ± 0.57	91.42 ± 0.61	93.68 ± 0.49	91.90–92.38
Multimodal pre-trained fusion[16]	90.36 ± 0.66	89.42 ± 0.71	91.82 ± 0.58	90.09–90.63
LLM-MalDetect[44]	91.34 ± 0.59	90.36 ± 0.64	93.04 ± 0.51	91.10–91.58
B. Proposed framework	**99.26 ± 0.18**	**99.20 ± 0.21**	**99.66 ± 0.12**	**99.19–99.33**

Comparison	Mean F1 difference (percentage points)	95% CI	Test statistic	Holm-adjusted p	Cohen's d_z
Proposed vs. Cyber code intelligence[3]	6.46	6.20–6.72	$t(24) = 51.27$	< 0.001	10.25
Proposed vs. AI-based malware analysis[8]	7.78	7.48–8.08	$t(24) = 54.02$	< 0.001	10.80
Proposed vs. Multimodal pre-trained fusion[16]	9.78	9.43–10.13	$t(24) = 57.93$	< 0.001	11.59
Proposed vs. LLM-MalDetect[44]	8.84	8.52–9.16	$t(24) = 56.91$	< 0.001	11.38

Table 9 Generalization, partition-integrity audit, and negative controls A: Generalization under distribution shift. B: Leakage audit and negative controls.

A. Evaluation protocol	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)
Group-aware controlled split	99.26	99.16	99.30	99.20	99.66
Temporally disjoint test	95.18	94.86	94.57	94.71	97.08
Family-disjoint test	92.34	91.82	90.96	91.37	94.65
B. Audit item	Detected before grouping	Remaining after grouping	Final result		
Exact binary duplicates across partitions	1,284	0	Passed		
Near-duplicate evidence groups across partitions	736	0	Passed		
Repeated episode identifiers across partitions	219	0	Passed		
Repeated source identifiers across partitions	148	0	Passed		
Family overlap in family-disjoint testing	63 groups	0	Passed		
Label-permutation accuracy	—	20.41 ± 0.73%	Near chance level		
Identifier-only classification accuracy	—	21.08 ± 0.81%	Near chance level		

Table 10 Component and alternative-design ablation (%).

Configuration	Classification accuracy	Alignment accuracy	Reconstruction accuracy	Attribution accuracy	Reliability
Without CTEEN	95.8	90.7	91.8	93.1	92.7
Without NSFRH	96.3	94.6	93.4	94.5	94.0
Without GASRE	97.0	97.8	89.2	95.8	95.4
Without CMES	97.8	98.1	97.1	96.9	96.8
Without CTAECO	98.1	98.3	97.5	92.4	95.9
Dynamic time warping instead of optimal transport	97.4	94.8	95.3	96.1	95.6
Pairwise graph instead of NSFRH hypergraph	97.1	96.2	94.6	95.2	95.0
Unconstrained reconstruction	97.6	97.9	92.1	96.0	94.8
Uncalibrated attribution	98.7	98.5	98.2	96.3	96.1
Complete framework	**99.3**	**98.9**	**98.9**	**99.2**	**99.0**

assignment reduced every prohibited overlap to zero. In the family-disjoint evaluation, 63 initially overlapping groups were reassigned before training, leaving no target-family overlap.

The label-permutation experiment achieved 20.41 ± 0.73% accuracy, while identifier-only classification achieved 21.08 ± 0.81%. Both values were close to the 20% balanced chance level. These controls provided no indication that the primary results could be reproduced through randomized labels or repository and episode identifiers alone.

Component and alternative-design ablation

Table 10 presents the module-removal and alternative-design ablation results.

The complete-framework classification value in Table 10 is rounded to one decimal place; therefore, 99.3% is consistent with the more precise 99.26% average reported in Tables 5 and 8. Similarly, the 98.9% alignment value represents Full Multimodal Fusion, whereas 98.46% represents the mean across all five alignment settings.

Table 11 Computational-efficiency comparison.

Method	Parameters (M)	Training time (h)	Peak memory (GB)	Latency per episode (ms)	Throughput (episodes/s)
Cyber code intelligence[3]	42.6	3.2	6.8	31.4	31.8
AI-based malware analysis[8]	35.8	2.7	5.9	27.6	36.2
Multimodal pre-trained fusion[16]	68.4	4.9	8.7	44.8	22.3
LLM-MalDetect[44]	124.7	6.8	11.6	61.5	16.3
Proposed framework	**96.3**	**7.4**	**12.8**	**73.2**	**13.7**

Removing CTEEN reduced classification accuracy from 99.3% to 95.8%, alignment accuracy from 98.9% to 90.7%, and reliability from 99.0% to 92.7%. This result supports the role of temporal-semantic correspondence before higher-order evidence reasoning.

Removing NSFRH reduced reconstruction accuracy to 93.4%, candidate-category attribution accuracy to 94.5%, and reliability to 94.0%. The result indicates that higher-order relationships contributed to both attack-stage interpretation and candidate differentiation.

Removing GASRE produced the largest reduction in reconstruction accuracy, from 98.9% to 89.2%. Removing CMES primarily reduced forecast-related reliability, while removing CTAECO produced the greatest decline in attribution accuracy, from 99.2% to 92.4%.

Replacing optimal transport with dynamic time warping reduced alignment accuracy from 98.9% to 94.8%. The reduction supports the use of soft many-to-many correspondence for asynchronously observed evidence. Replacing NSFRH with pairwise graph reasoning reduced reconstruction and attribution accuracy to 94.6% and 95.2%, respectively, supporting the use of higher-order relations.

Unconstrained reconstruction reduced reconstruction accuracy to 92.1% and reliability to 94.8%, demonstrating the contribution of the evidence-consistency term. The uncalibrated attribution alternative retained comparatively strong classification performance but reduced attribution accuracy to 96.3% and reliability to 96.1%.

The calibrated CTAECO configuration achieved an expected calibration error of 1.8% and a Brier score of 0.024, compared with 7.6% and 0.081 for the uncalibrated alternative. These results demonstrate that calibration improved confidence quality rather than merely classification accuracy.

Computational efficiency

Table 11 compares model size, training time, memory consumption, episode-level latency, and throughput.

The proposed framework required 96.3 million trainable parameters, 7.4 h of training, and 12.8 GB of peak memory. Its mean latency was 73.2 ms per complete multimodal episode, corresponding to 13.7 episodes/s.

This computational cost was higher than that of the classification-oriented baselines because the proposed framework additionally performed temporal alignment, hypergraph reasoning, evidence-constrained reconstruction, counterfactual simulation, and confidence aggregation. LLM-MalDetect[44] contained more parameters but produced lower latency because its adapted implementation did not execute the complete five-module forensic pipeline.

The proposed framework therefore exchanges additional computational cost for broader forensic functionality. Its episode-level latency supports analyst-assisted investigation, although deployment on highly constrained devices may require pruning, quantization, or intermediate-representation caching.

Controlled case illustration

A controlled financial-sector scenario was used to illustrate the structure of the final forensic report. The scenario was synthetic and was not presented as deployment at an actual financial institution. The illustration contained 5,000 network sessions, 1,200 malware files, 850 phishing emails, and 300 voice-based social-engineering

recordings, producing 7,350 evidence items. CTEEN aligned approximately 7,180 items under the controlled episode rules, corresponding to approximately 97.7% alignment. NSFRH represented the aligned evidence through relationships connecting phishing instructions, script execution, executable behaviour, and command-and-control communication. GASRE reconstructed benchmark-defined stages omitted from the controlled episode. CMES generated five counterfactual variations, each explicitly labelled as a prospective scenario rather than historical evidence. CTAECO produced a confidence-scored attribution hypothesis restricted to the candidate categories represented in the benchmark. This illustration demonstrates the format of the generated forensic report. It is not an independent held-out experiment and was excluded from the primary averages, confidence intervals, and statistical comparisons.

Discussion

The results indicate that the proposed framework achieved strong performance across classification, controlled cross-modal alignment, evidence-constrained reconstruction, candidate-category attribution, and forensic reliability. The improvement over Cyber Code Intelligence[3], AI-Based Malware Analysis[8], Multimodal Pre-trained Fusion[16], and LLM-MalDetect[44] was associated with the broader task coverage of the proposed framework. The baseline implementations primarily classified malware or fused selected representations. The proposed system additionally aligned asynchronous evidence, represented higher-order forensic relationships, reconstructed missing benchmark stages under evidence constraints, separated counterfactual projections from historical evidence, and calibrated candidate-category attribution confidence.

The high results require controlled interpretation. First, the integrated episodes were constructed using compatible threat categories, forensic-stage rules, and observable relationships. This setting is more structured than naturally co-collected operational incidents. Second, the source repositories contain recognizable malware and attack characteristics. Performance decreased under both temporal and family distribution shift. Third, classification, alignment, reconstruction, attribution, and reliability use different targets; numerical similarity among these tasks does not indicate that they measure the same property. The temporal- and family-disjoint experiments provide a more cautious assessment of generalization. The family-disjoint accuracy of 92.34% is substantially lower than the controlled-split accuracy of 99.26%, demonstrating that previously unseen families remain more difficult. Therefore, zero-day or unseen-family claims should be based on the family-disjoint result and not the controlled-split performance.

The leakage audit and negative controls provide evidence that duplicate overlap and identifier-based shortcut learning did not account for the reported performance. However, the evaluation remains limited to the specified public repositories, controlled correspondence rules, and candidate categories. The results support the internal effectiveness of the five-module architecture under the defined benchmark. They do not establish unrestricted real-world performance, courtroom admissibility, or definitive attribution to real threat actors. Validation using naturally co-collected external incidents remains necessary to establish broader operational applicability.

Conclusion, limitations, and future work

Conclusion

This study presented a Multimodal Generative Cyber Forensics Network integrating CTEEN, NSFRH, GASRE, CMES, and CTAECO for temporal-semantic evidence alignment, higher-order forensic reasoning, evidence-constrained attack-stage reconstruction, counterfactual threat analysis, and confidence-aware candidate-category attribution. The framework was evaluated using VirusShare, CIC-MalMem-2022, the Android Malware Corpus,

the Malware Traffic Repository, and an Integrated Controlled Multimodal Benchmark. Full audio–text–binary–traffic fusion was evaluated only on the integrated benchmark.

Under the group-aware controlled protocol, the framework achieved 99.26% classification accuracy, 99.20% F1-score, 99.66% AUC, 98.46% mean alignment accuracy, 98.68% reconstruction accuracy, 99.04% candidate-category attribution accuracy, and 99.0% overall reliability. Module and design-choice ablations confirmed the contributions of temporal alignment, hypergraph reasoning, evidence-constrained reconstruction, and confidence calibration. However, accuracy decreased to 95.18% under temporally disjoint testing and 92.34% under family-disjoint testing, demonstrating the greater difficulty of chronological drift and previously unseen families.

These findings support the effectiveness of the proposed framework within the controlled benchmark. They do not establish natural incident-level correspondence, definitive real-world threat-actor attribution, or universal forensic performance. Observed evidence, reconstructed stages, and counterfactual projections must therefore remain explicitly separated during interpretation.

Limitations

The integrated benchmark was constructed from independently collected repositories rather than naturally co-collected operational incidents. Performance may decline under missing modalities, encrypted traffic, unseen candidate categories, adversarial evidence, or substantial domain shift. Attribution remains limited to benchmark-defined candidate categories, while reconstructed and counterfactual outputs represent evidence-supported hypotheses rather than directly observed facts. The framework also introduces computational overhead, requiring 96.3 million parameters, 12.8 GB peak memory, and 73.2 ms per multimodal episode.

Future work

Future work will evaluate the framework using naturally co-collected incident-level evidence and broader cross-organizational, temporal, family-disjoint, and open-set settings. Further research will investigate provenance-preserving federated learning, evidence-grounded language-model assistance, adversarial robustness, and uncertainty calibration under distribution shift. Pruning, quantization, sparse hypergraph computation, and knowledge distillation will also be explored to reduce memory and inference cost while preserving forensic reliability.

Author contributions Rijvan Beg was involved in experimentation, coding, and writing the initial draft. R.K Pateriya, Deepak Singh Tomar, and Ghanshyam Raghuwanshi, Surendra Solanki were involved in ideation, Analysis, supervision, and preparing the final draft.

Data availability The datasets used in this study are publicly available from their respective official sources. The VirusShare dataset is available at https://virusshare.com (registration required). The CIC-MalMem-2022 dataset can be accessed through the Canadian Institute for Cybersecurity (CIC). The Android malware dataset and Malware Traffic Repository are available from their respective official repositories. Any additional processed data generated during this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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